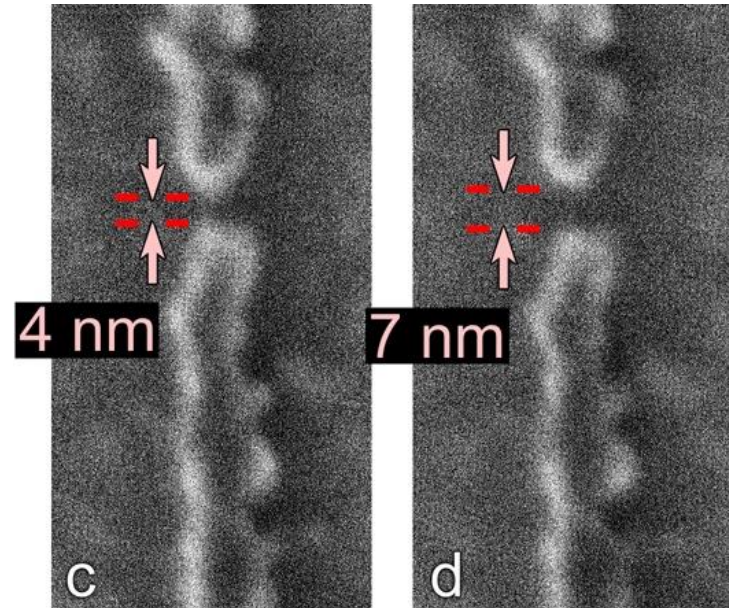




68320 Nanofabrication and and Nanocharacterization Techniques

Demonstrator
Hossein Alijani

Lecturer & Subject Coordinator
Milos Toth



Tutorial 1: Nano-scale imaging basics
magnification, resolution, signal-to-noise-ratio and image contrast

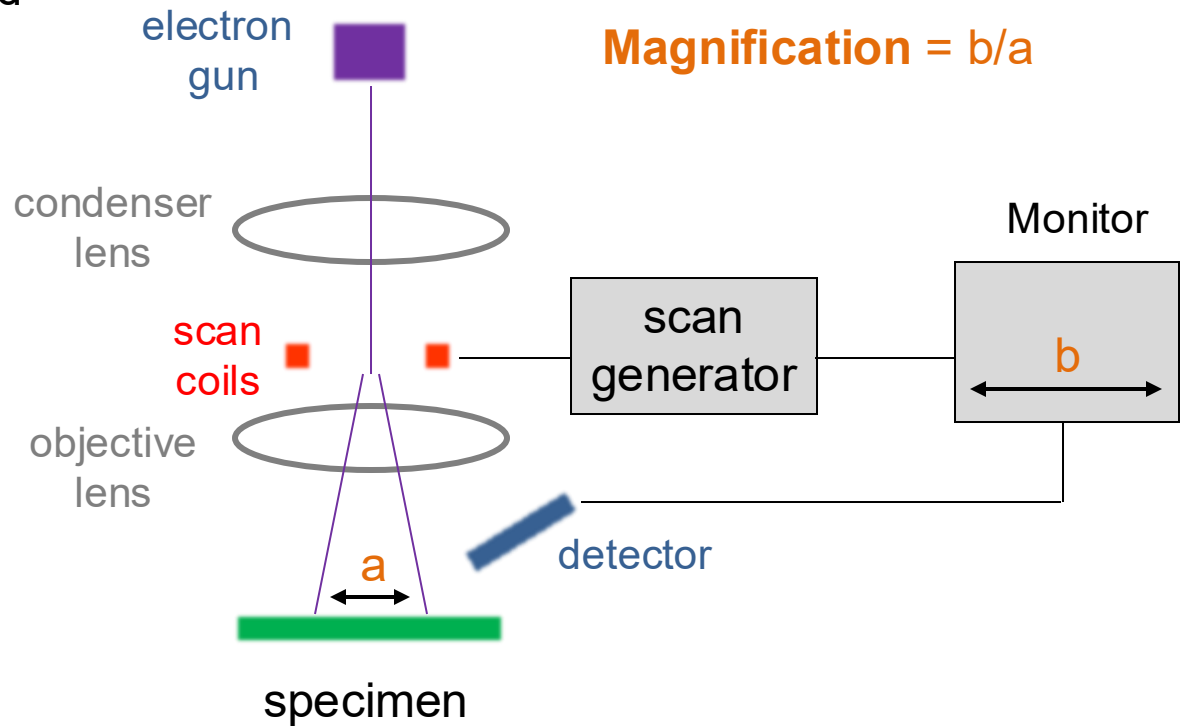
SEM basics & magnification

all SEMs operate by irradiating the specimen with a finely focussed beam of electrons, producing emission signals (SE, BSE, X-ray, etc.) that can be used to form an image

Electron Gun: used to produce electrons, pre-focus beam and accelerate to 1 - 30 keV

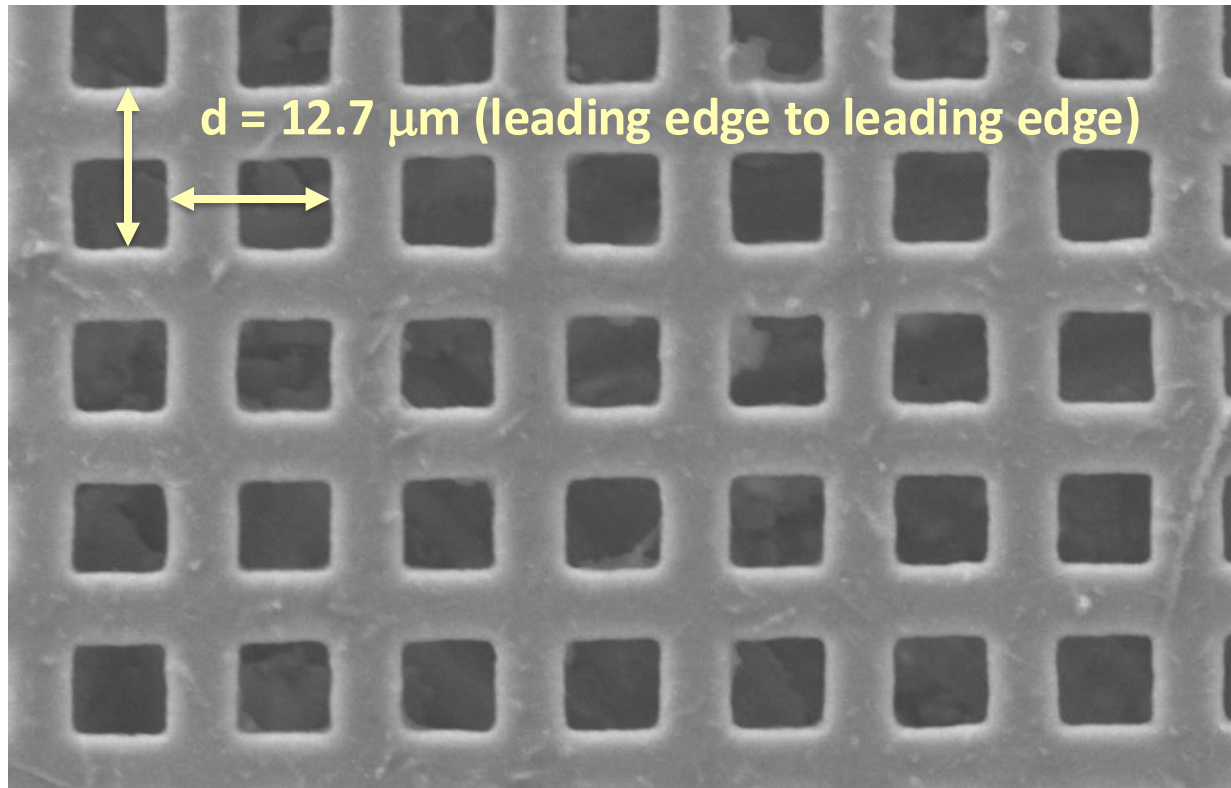
Electron Lens: used to provide a focussed, de-magnified spot at the specimen plane

Imaging System: used to scan the beam, achieve magnification, and produce image contrast

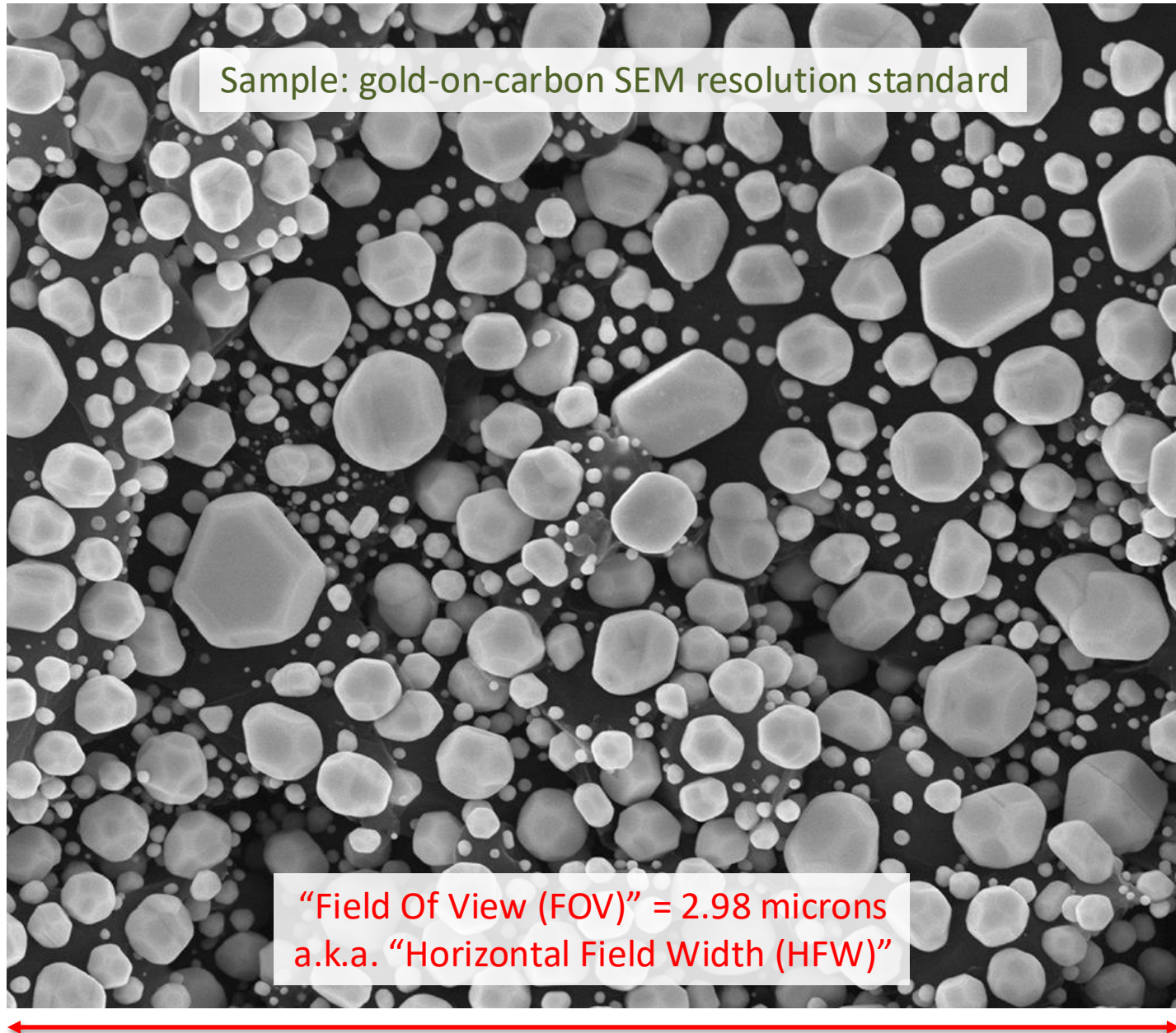


Magnification calibration

SEM magnification must be calibrated using an image of a **calibration standard**

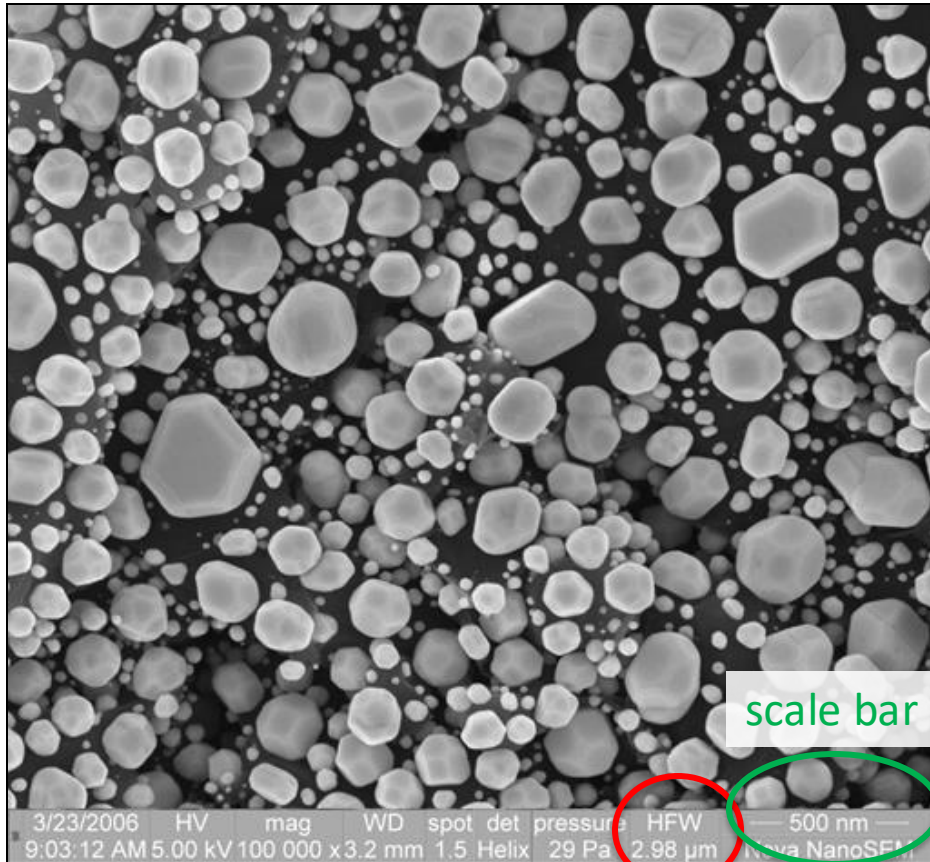


Field width, magnification & resolution



Field width, magnification & resolution

“Field Of View (FOV)” = 2.98 microns
a.k.a. “Horizontal Field Width (HFW)”



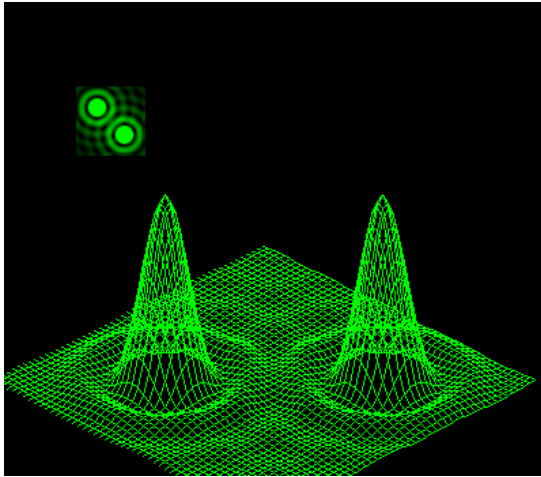
width of this image = 3584 pixels
FOV = 2.98 μm
hence, 1 pixel = 0.831 nm

SEM images often only show a scale bar
here, scale bar = 500 nm (601 pixels)

What is the resolution of this image?

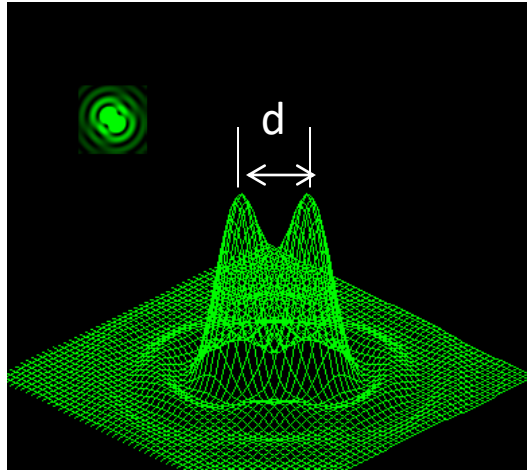
What is resolution?

resolved

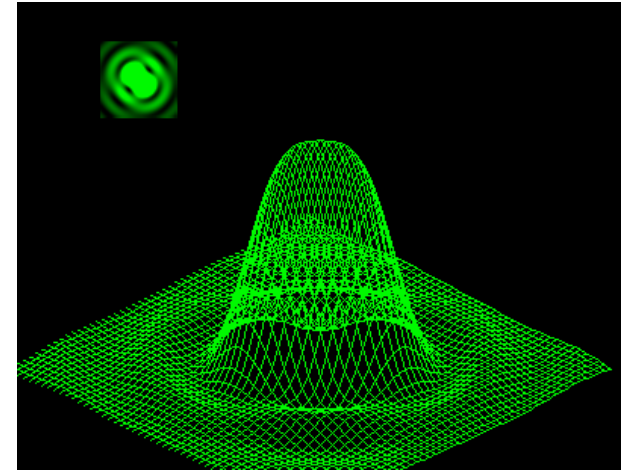


diffraction-limited image of 2 features much smaller than the resolution

resolved



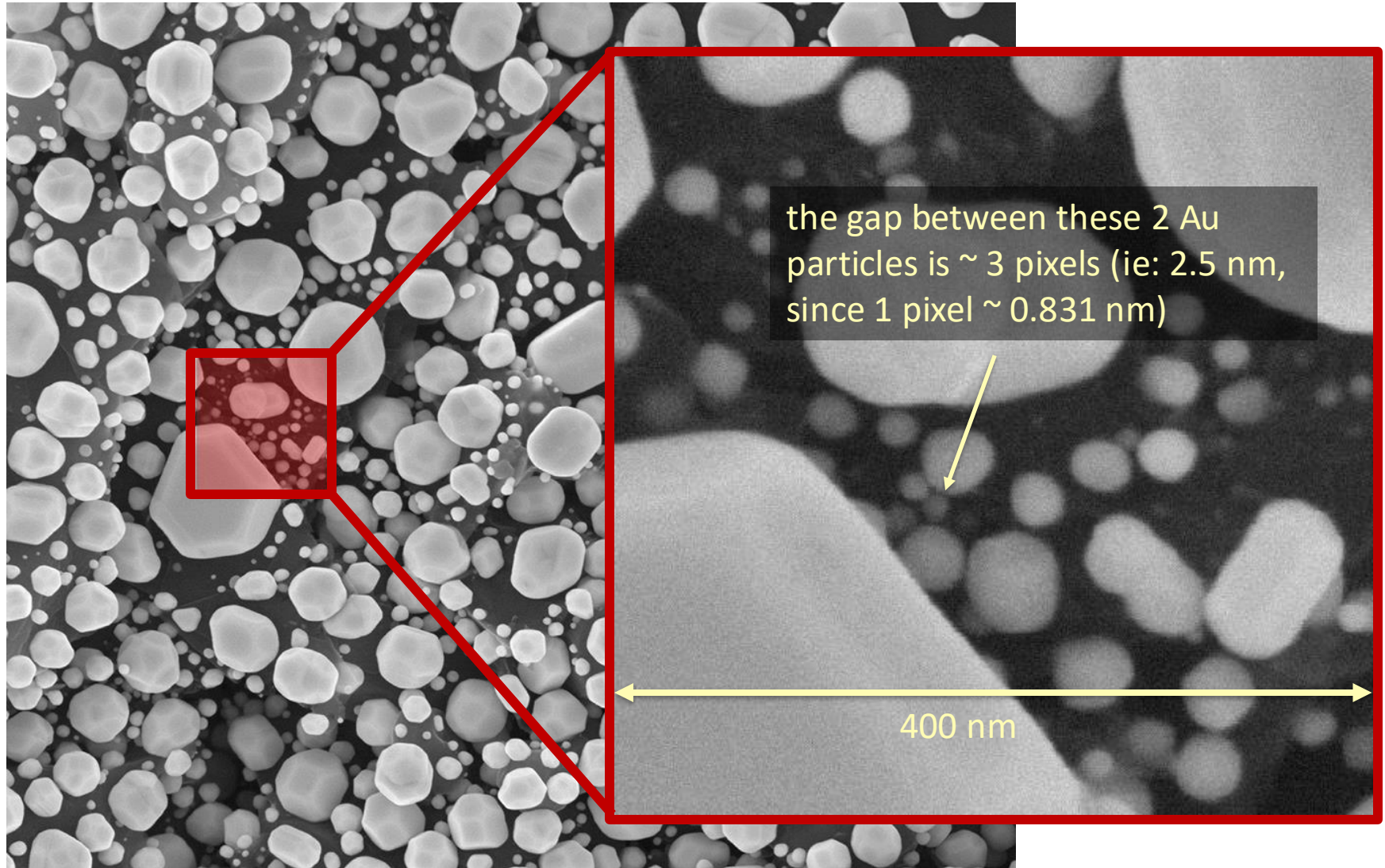
unresolved



is this an image of 1 broad feature or 2 small features?

spatial resolution: smallest distance between two features that can be distinguished as two separate entities

How can we measure SEM resolution?



... in practice, people often simply measure the shortest distance between 2 Au particles in an image of a “gold-on-carbon resolution standard”

A more sophisticated approach to
measuring resolution and other
metrics of image quality

Reading material

The following paper is in Canvas:

Canvas/Modules/Tutorials/Tute 1/06. SEM Resolution.pdf

This paper was published here: Proc. SPIE 3998, Metrology, Inspection, and Process Control for Microlithography XIV, (2 June 2000); doi: 10.1117/12.386444

Metrics of resolution and performance for CD-SEMs

David C Joy^{a,b}, Yeong-Uk Ko^a, and Justin J Hwu^a

^aEM Facility, University of Tennessee, Knoxville, TN 37996

^bOak Ridge National Laboratory, Oak Ridge, TN 37831

This program is now known as "ImageJ" or "FIJI". It's implemented in Java and runs on both Macs and Windows machines. It's freeware & can be downloaded from the web

ABSTRACT

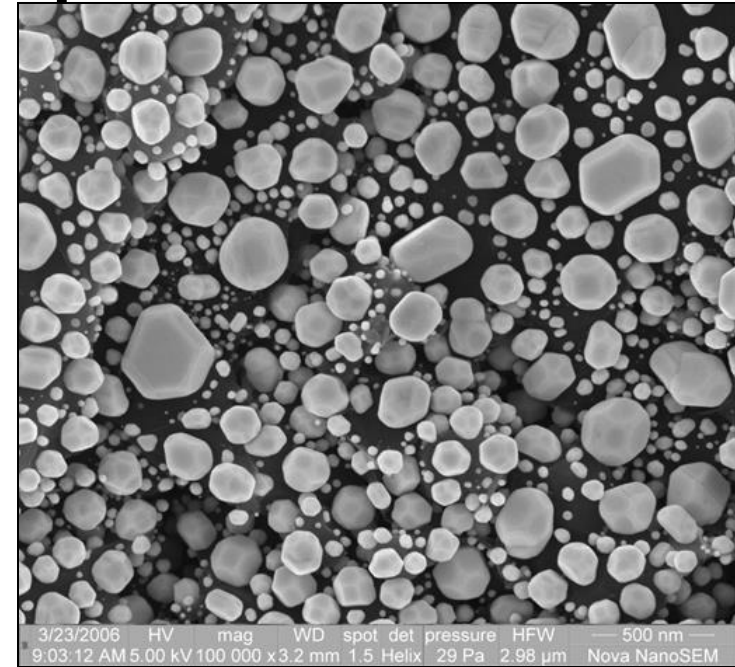
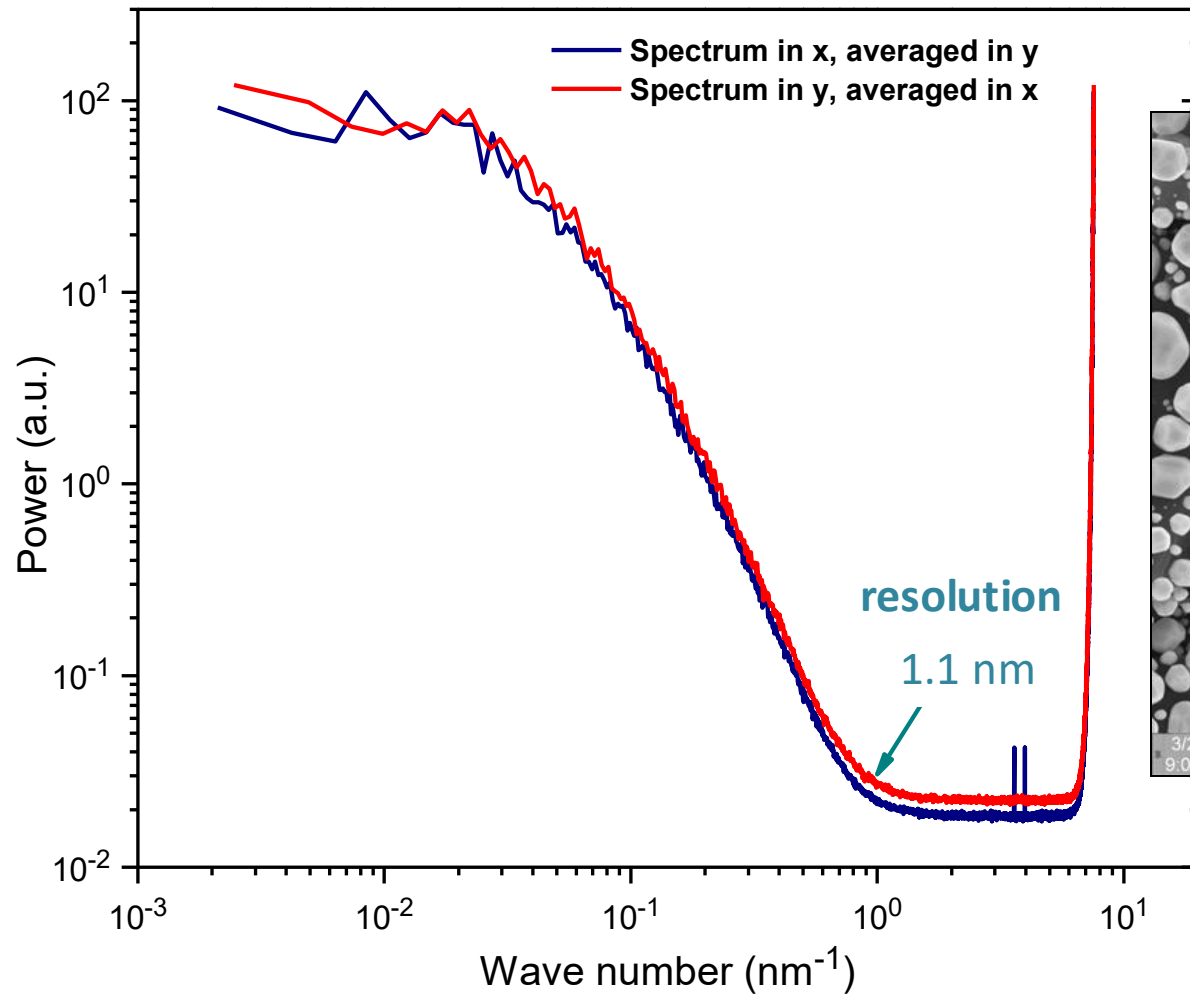
The performance of scanning electron beam instruments such as CD-SEMs can be defined in terms of parameters such as the beam probe size, the spatial resolution, and the signal to noise ratio of the image. A knowledge of these quantities is important in verifying the fact that an instrument meets its specification, and subsequently for tracking and optimizing its performance during use. Analytical methods based on the power spectrum (2-D Fourier transform analysis) of images are now beginning to be used for these purposes but care must be exercised to ensure reliable and meaningful results. Two new methods are suggested which can offer more detailed information about the microscope performance while avoiding the pitfalls of the simpler technique. Code implementing these tests, written as a plug-in macro for the well known NIH Image program, is available on-line.

Keywords: scanning electron microscope, resolution, signal to noise, probe size, Fourier transform

1. INTRODUCTION

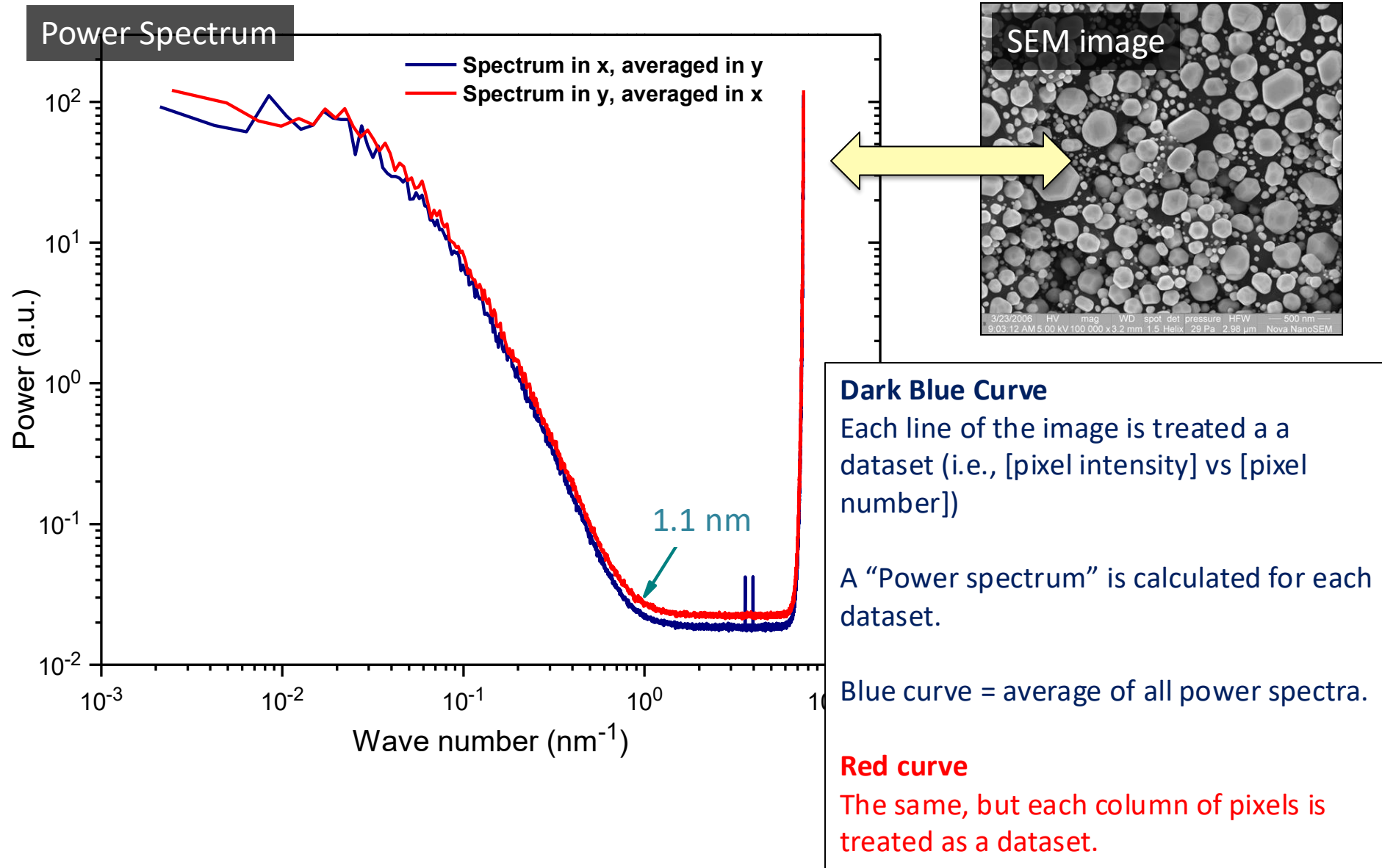
Scanning electron microscopes designed for the critical dimension metrology (CD-SEMs) must achieve a resolution approaching 1nm for the current technology nodes¹. Verifying that the tool is reaching the required level of performance when it is delivered, and subsequently monitoring its performance during use, requires that tests be available which can quantify the parameters such as probe size, resolution, the signal to noise ratio of the image under standard operating conditions etc., which define the performance level. In addition many electron beam tools are now used in a semi-automatic fashion with little or no operator supervision. There is thus a need for techniques which can monitor the performance of the tool to ensure that it remains within acceptable limits. Ideally such tests should be capable of being performed

Power spectrum of an image



Understanding a power spectrum

essentially a measure of image **signal level** vs “wave number” (or “spatial frequency”)



Understanding a power spectrum

Revision: Fourier Transform & Power Spectrum – review your notes from prior subjects and/or look it up online (start with Wikipedia)

Fourier Transform: a mathematical technique that is used to transform a function of time or space, into a function of frequency.

The Fourier Transform is a complex function. It represents both the magnitude and phase offset of each frequency (sinusoid) making up the signal.

Power Spectrum: the square of the absolute value of the Fourier Transform:

Used to identify the frequency components of a signal or to find the dominant frequencies in a signal.

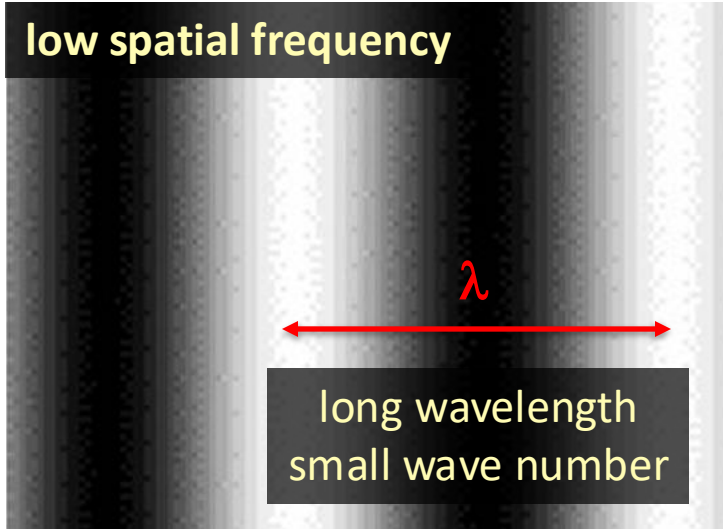
Here we use it to identify spatial frequencies in an image.

Understanding a power spectrum

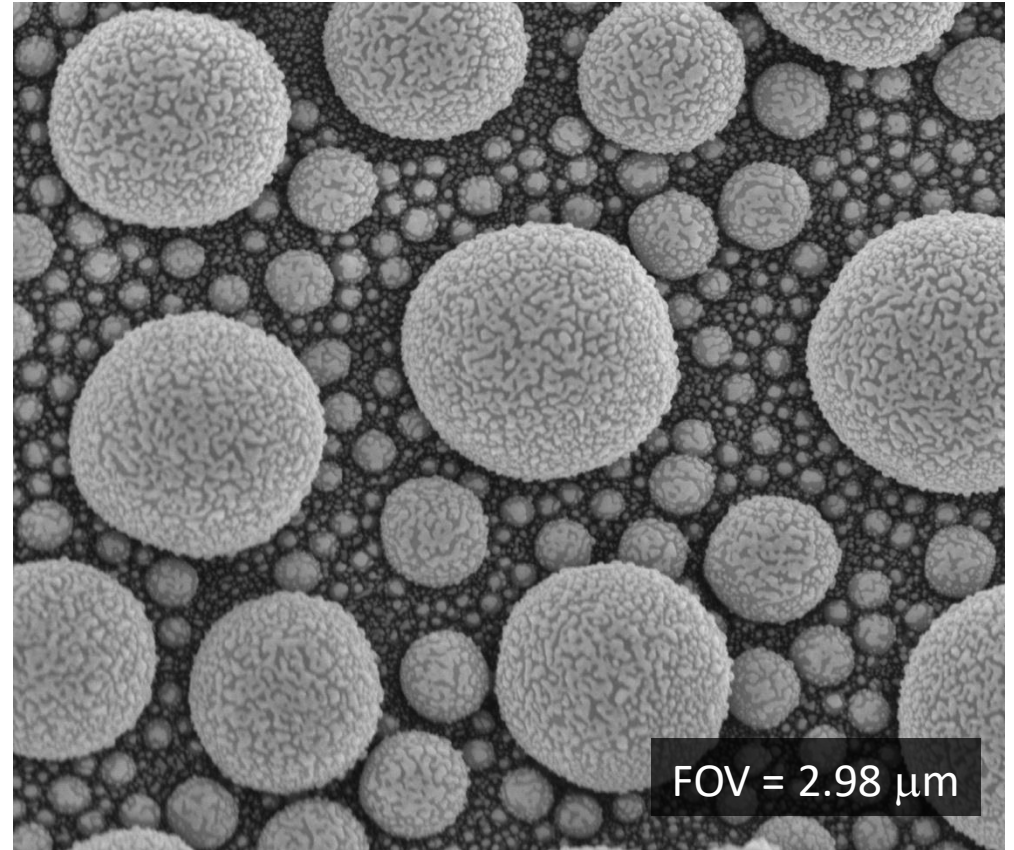
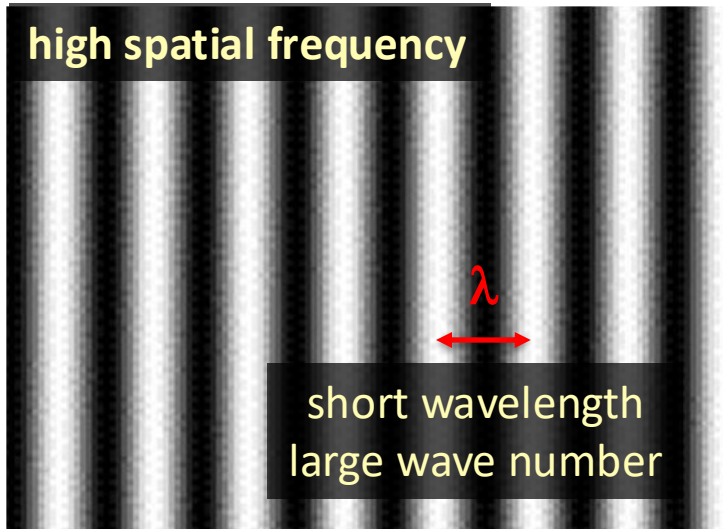
Period (i.e., wavelength): λ

Wavenumber: $k = 1/\lambda$

low spatial frequency



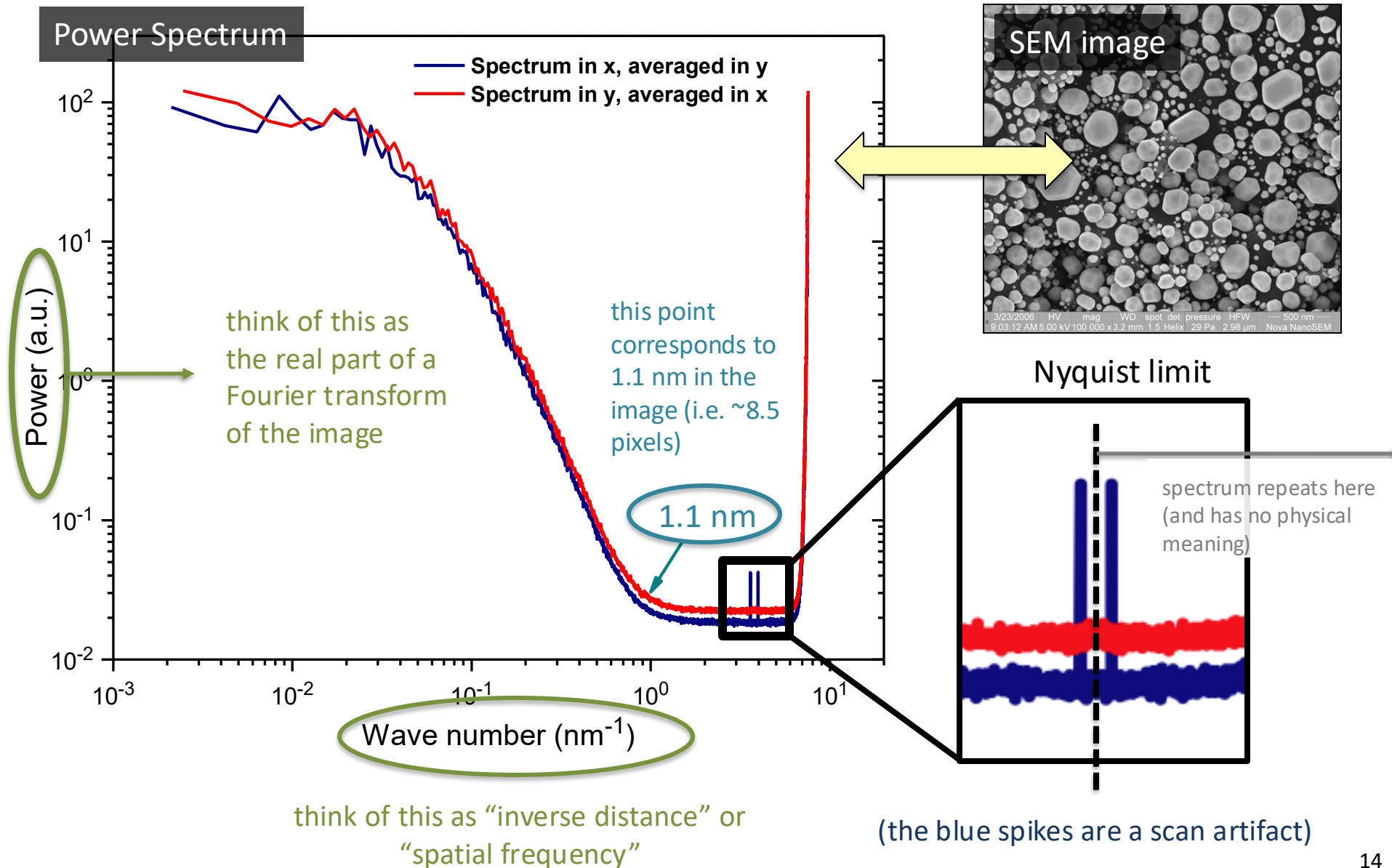
high spatial frequency



SEM image with a wide range of spatial frequencies

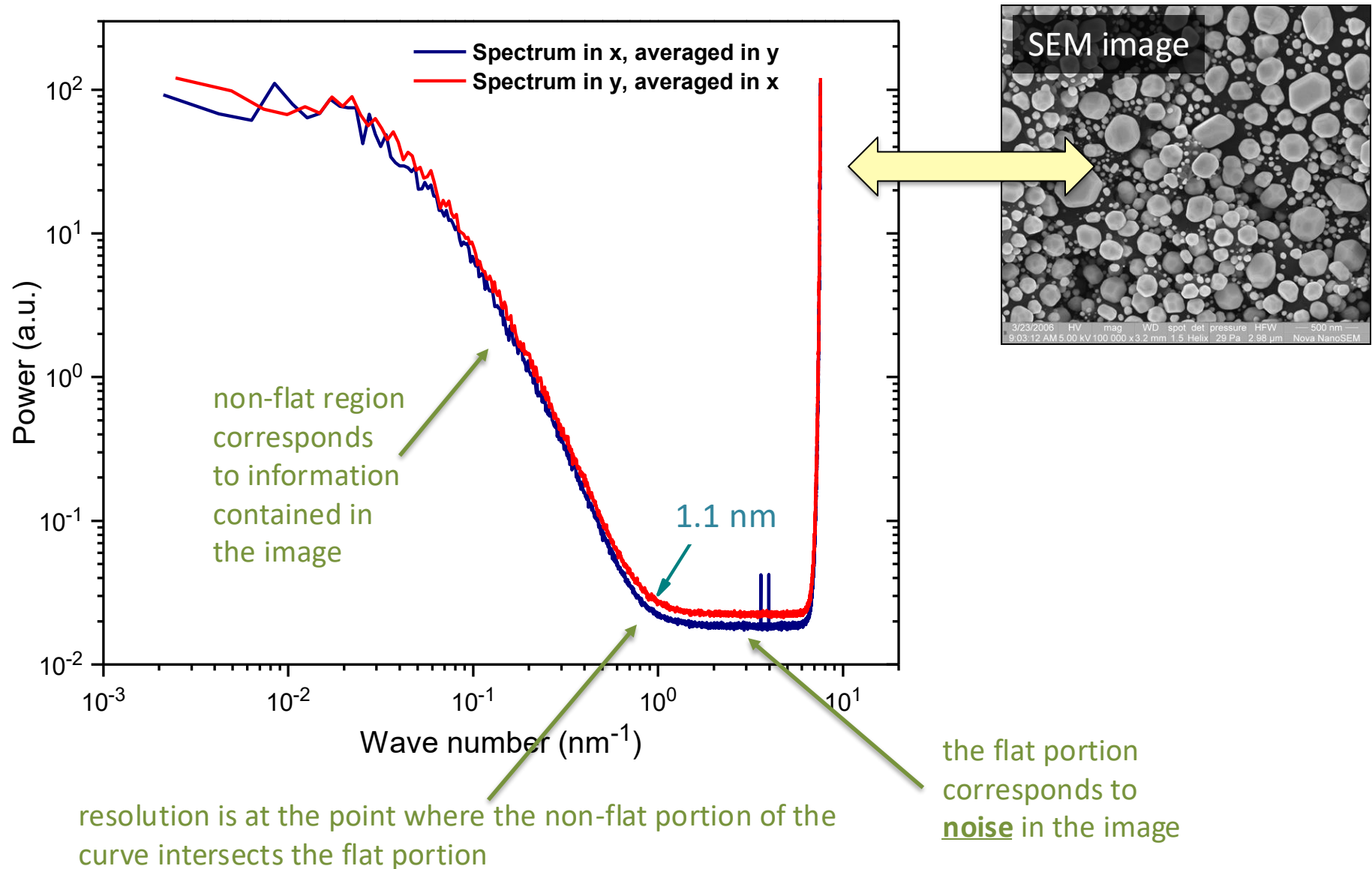
Understanding a power spectrum

essentially a measure of image **signal level** vs “wave number” (or “spatial frequency”)

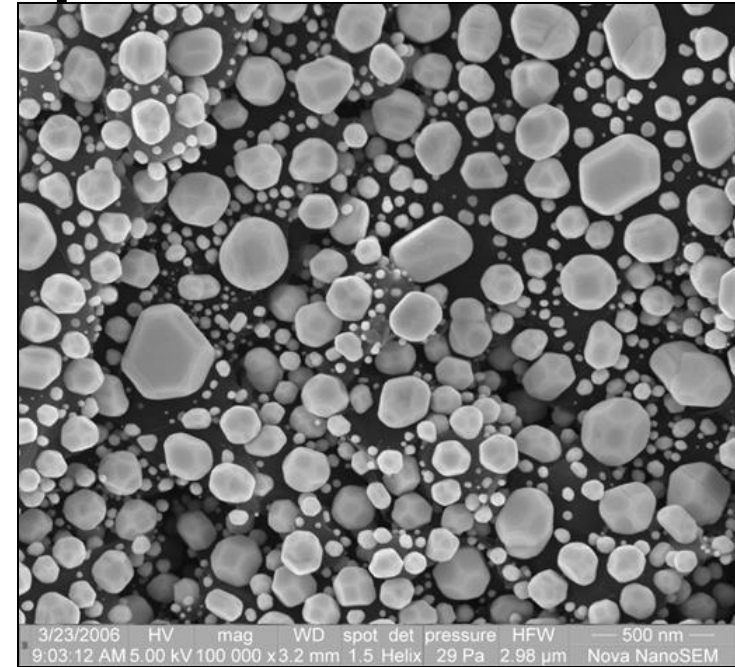
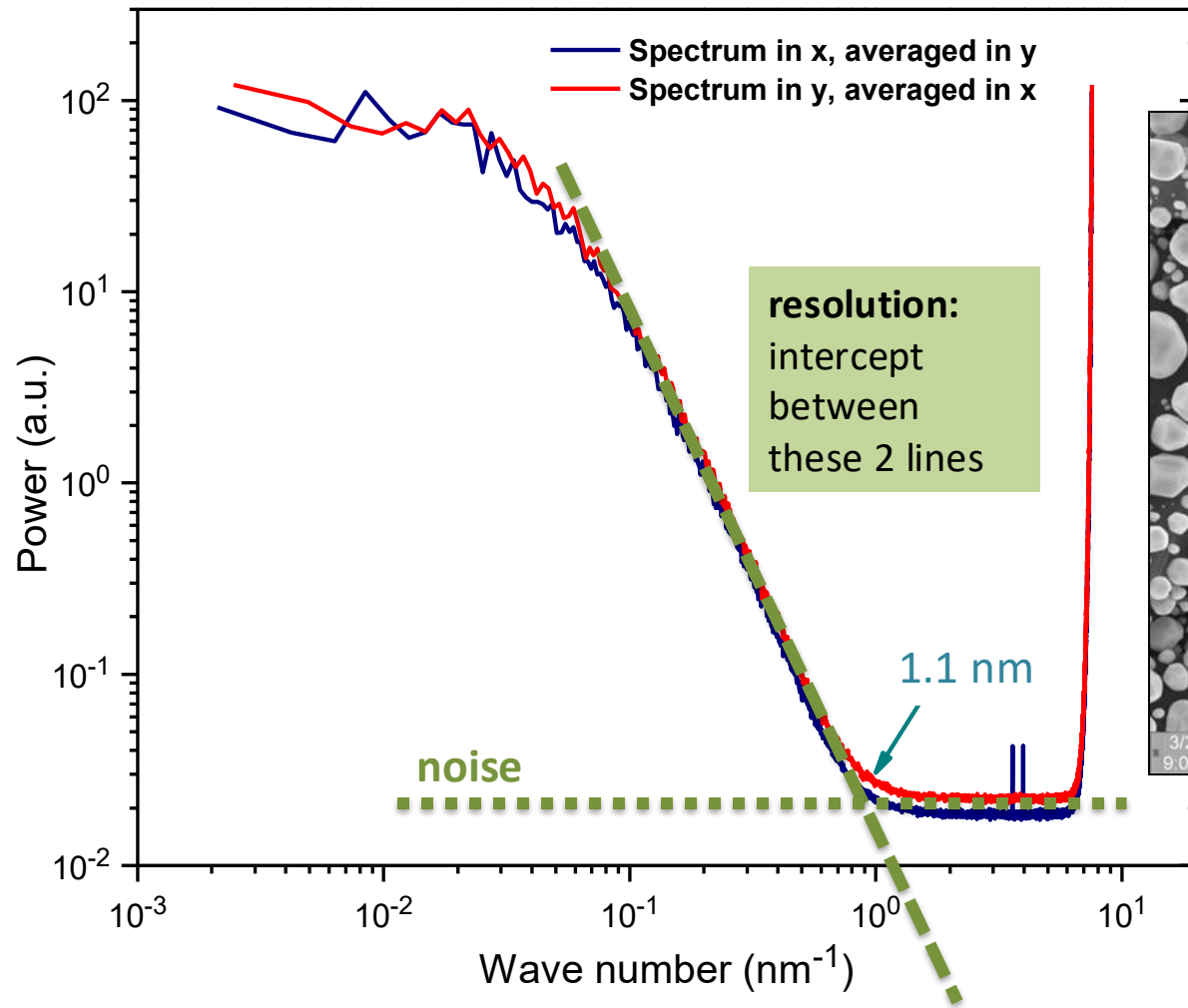


Understanding a power spectrum

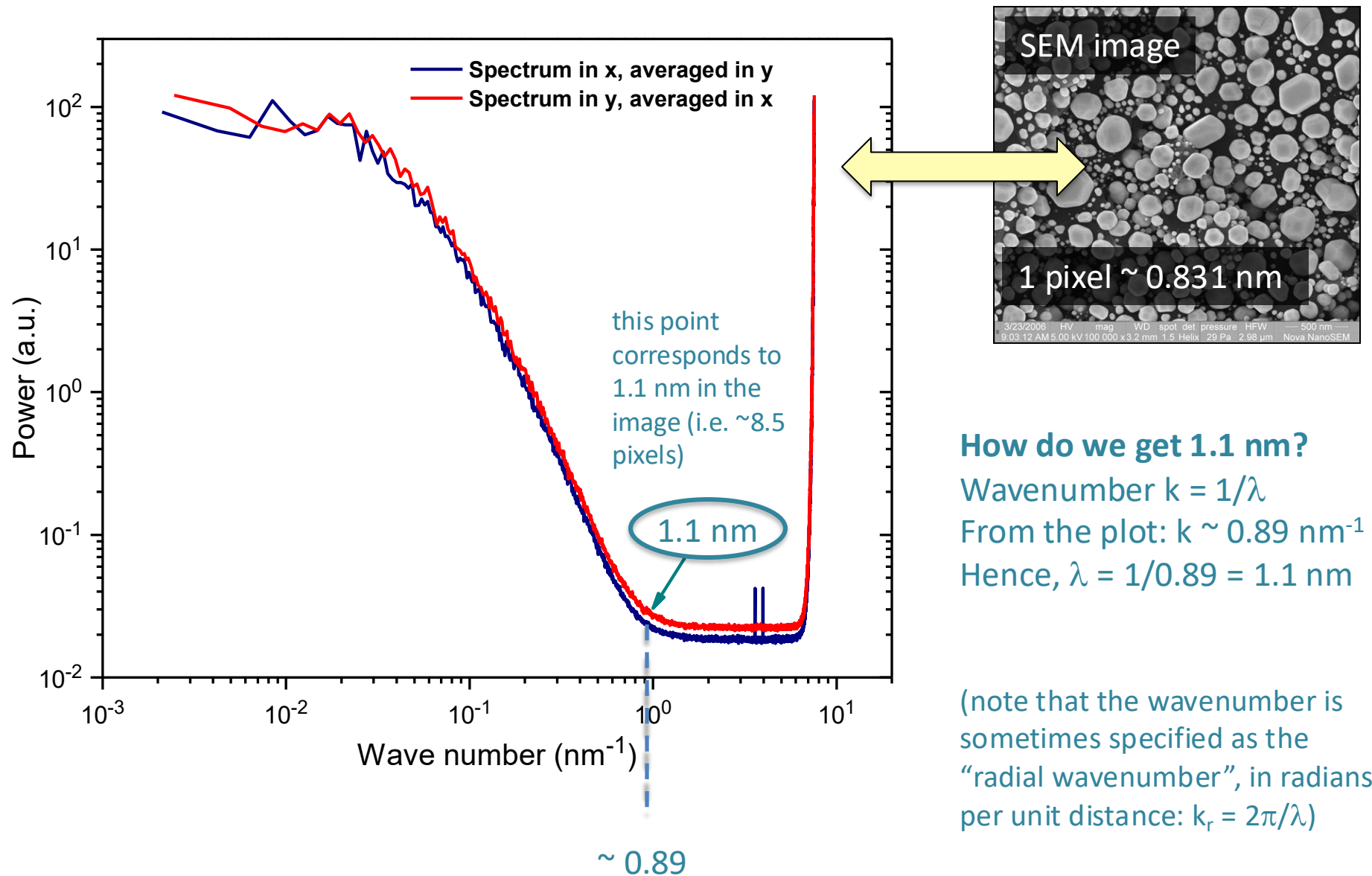
essentially a measure of image **signal level** vs “wave number” (or “spatial frequency”)



Measuring resolution

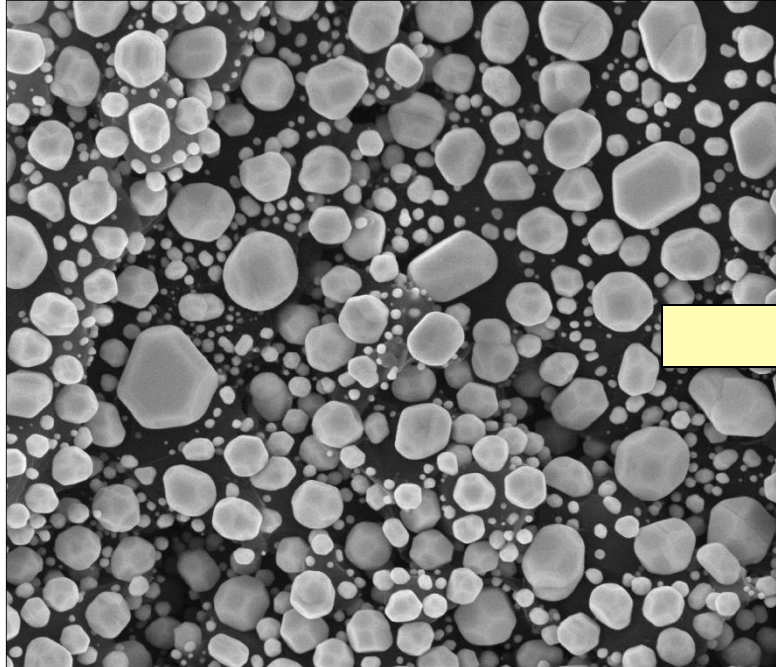


Understanding a power spectrum



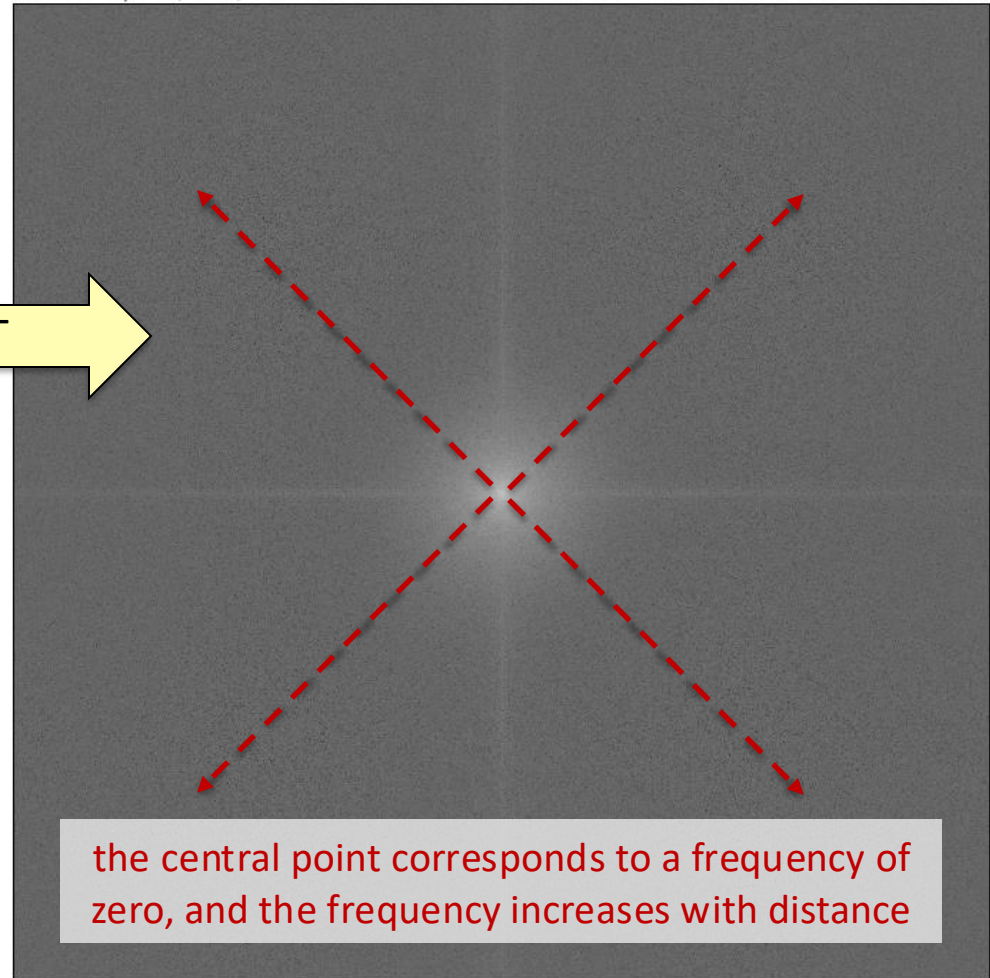
Radial Fourier Transform of an image

image



FFT

Fast Fourier Transform (FFT)



Check out **FIJI** – a free Image analysis/processing program written in Java:
<https://fiji.sc/>

FIJI is a release of the program **ImageJ** that includes various image processing plugins.

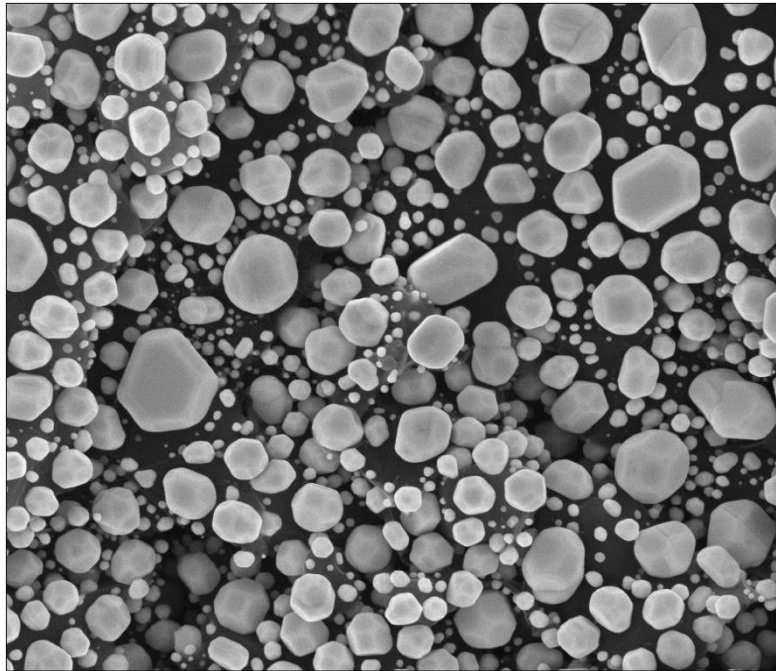
Installed on PCs in computer lab:

Windows → Start → FIJI → Launch_FIJI

Open image [balls1-2048x2048.tif]; Menu → Process → FFT → FFT

Image resolution measurement in 2D

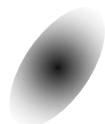
image



- in 2D, resolution is defined by a circle
- an ellipse indicates an astigmatic image (i.e., the electron beam is elliptical rather than circular and the resolution is different in different directions)

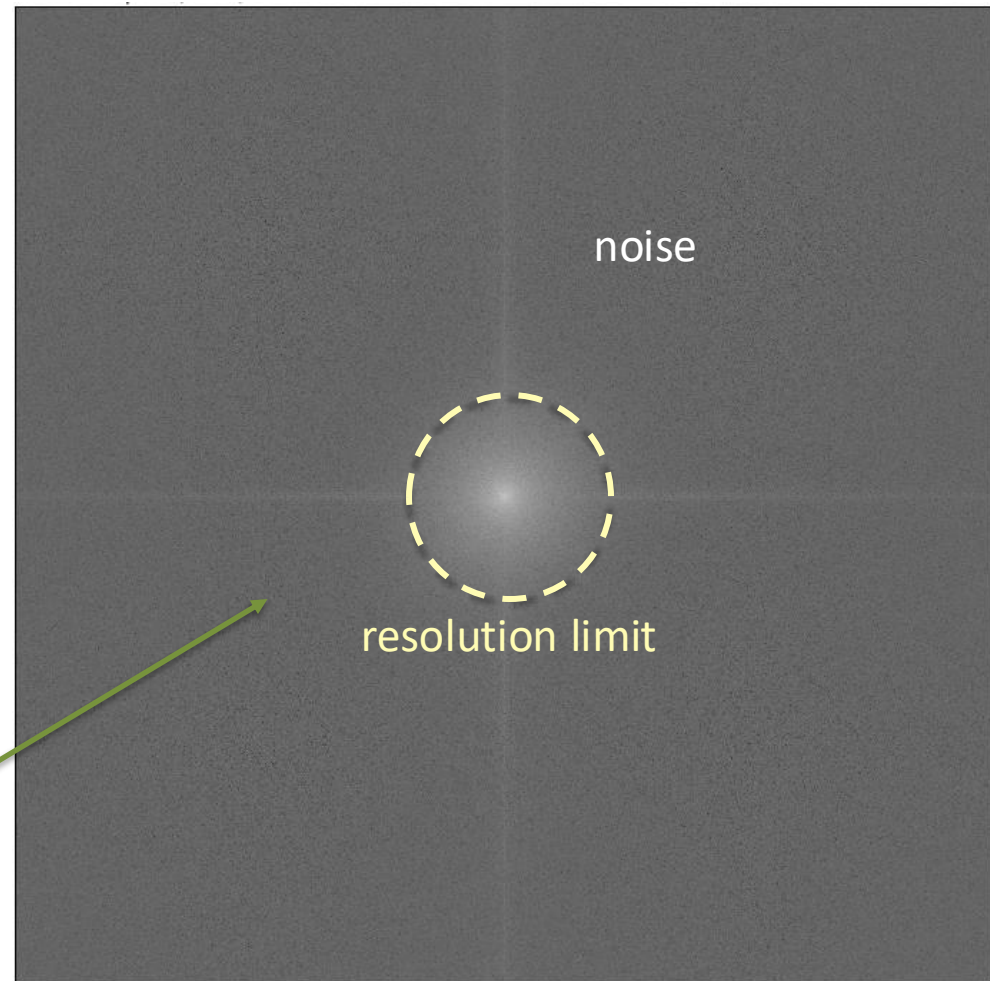


ideal beam



astigmatic beam

Fast Fourier Transform (FFT)



as in 1D, the challenge here is how to decide where the circle should be drawn (ie: what is the resolution???)

Autocorrelation: A robust method objective measurement of resolution

Cross-correlation function (CCF): a measure of the similarity of two datasets (i.e., images) as a function of the displacement of one relative to the other. A standard method of estimating the degree to which two datasets are correlated.

The CCF of 2 datasets denoted by $f(t)$ and $g(t)$ is the convolution of $f^*(-t)$ and $g(t)$

Autocorrelation function: the special case of a CCF applied to 2 copies of the same image.

CCF: General definition for discrete functions (datasets) f & g in 1D

cross-correlation
operator

denotes a complex
conjugate

$$(f \star g)[n] \stackrel{\text{def}}{=} \sum_{m=-\infty}^{\infty} f^*[m] g[m+n]$$

the complex conjugate of a complex number is the number with an equal real part and an imaginary part equal in magnitude but opposite in sign (not relevant to SEM images which are made up of real numbers)

Homework: Look up the relationships between a Fourier Transform, Convolution, Correlation and Autocorrelation

ImageJ software

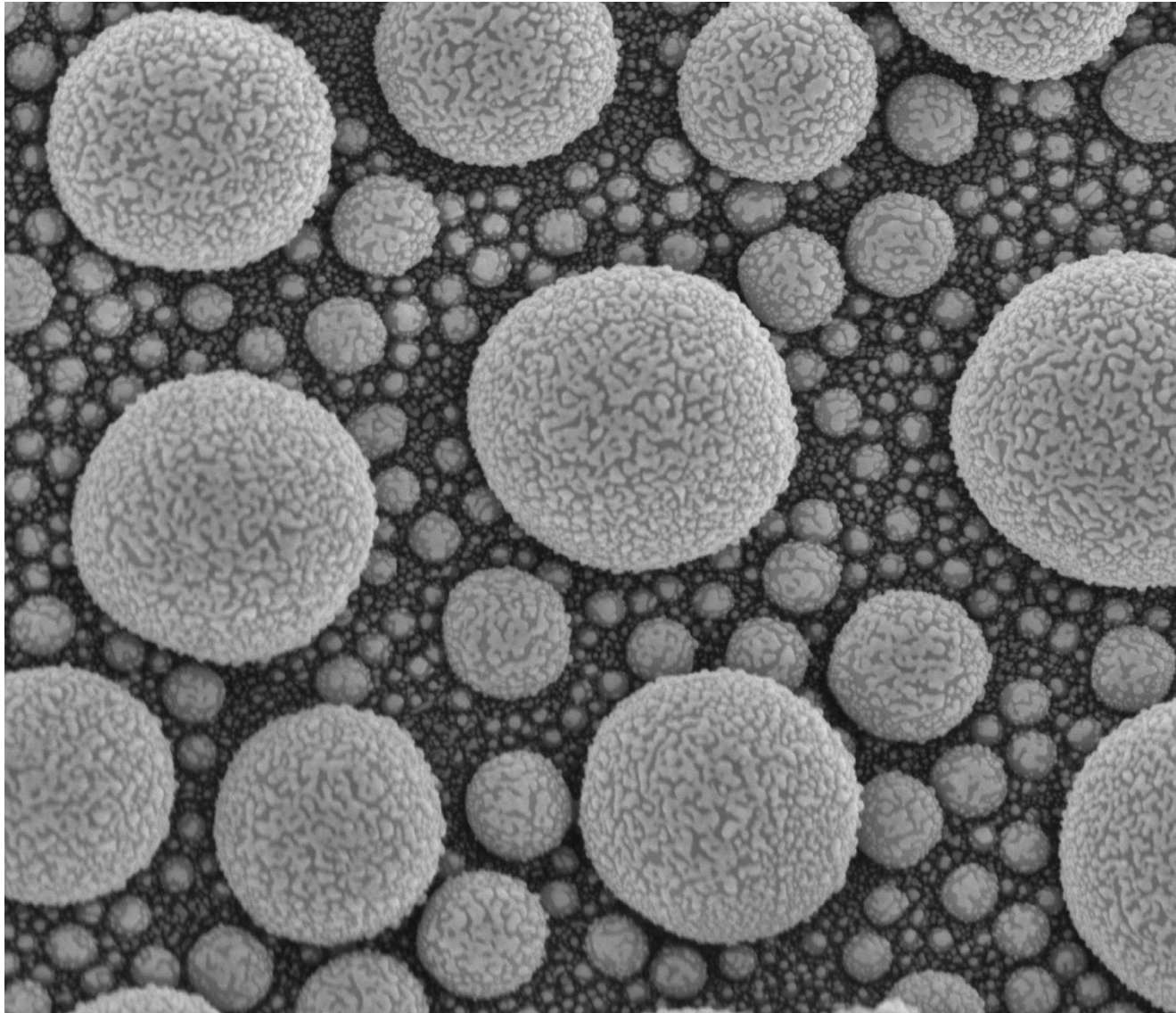
ImageJ was used to perform the image analysis on the following slides.

All the images in this tutorial are available in:
Canvas/Modules/Tutorials/Tute 1

FIJI is a release of the program **ImageJ** that
includes various image processing plugins:
<https://fiji.sc/>

It is installed on PCs in the computer lab:
Windows → Start → FIJI → Launch FIJI

Autocorrelation in ImageJ



**STEP 1: Use an SEM
to collect an image**

Image of sample

FOV = 2980 nm

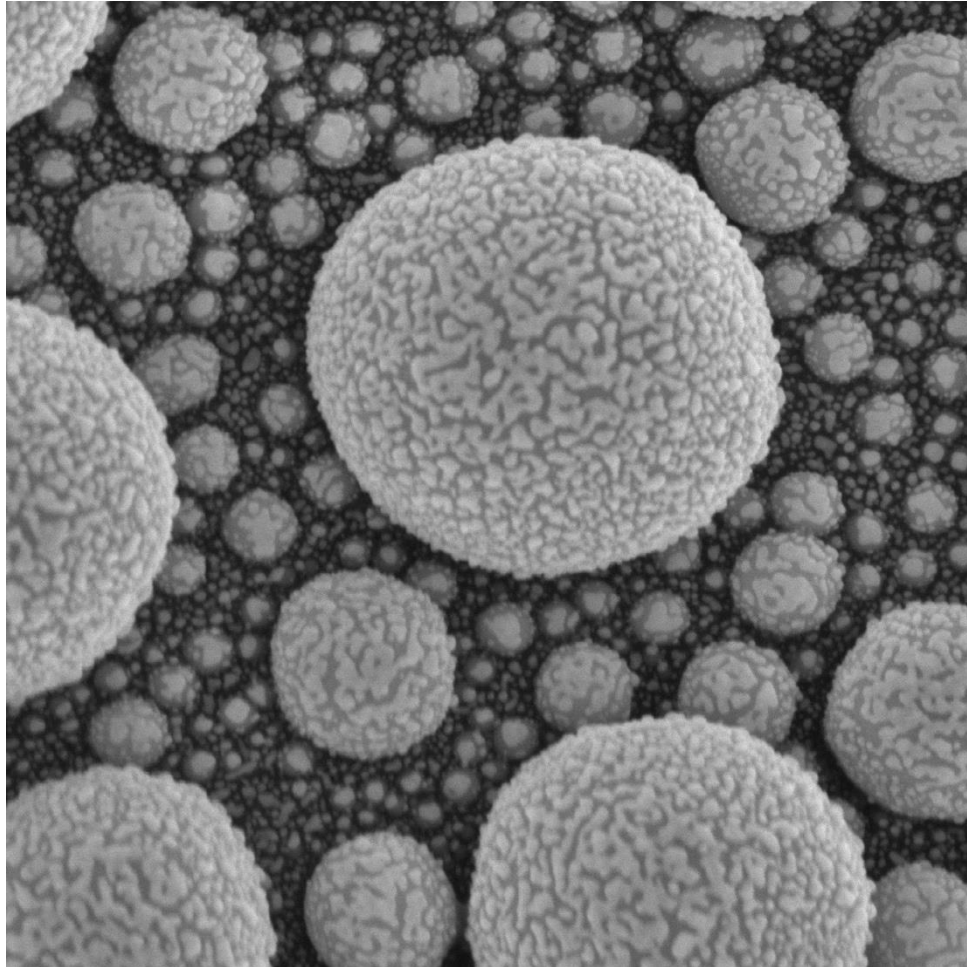
FOV = 3584 pixels

ie: 1 pixel $\sim$ 0.8315 nm

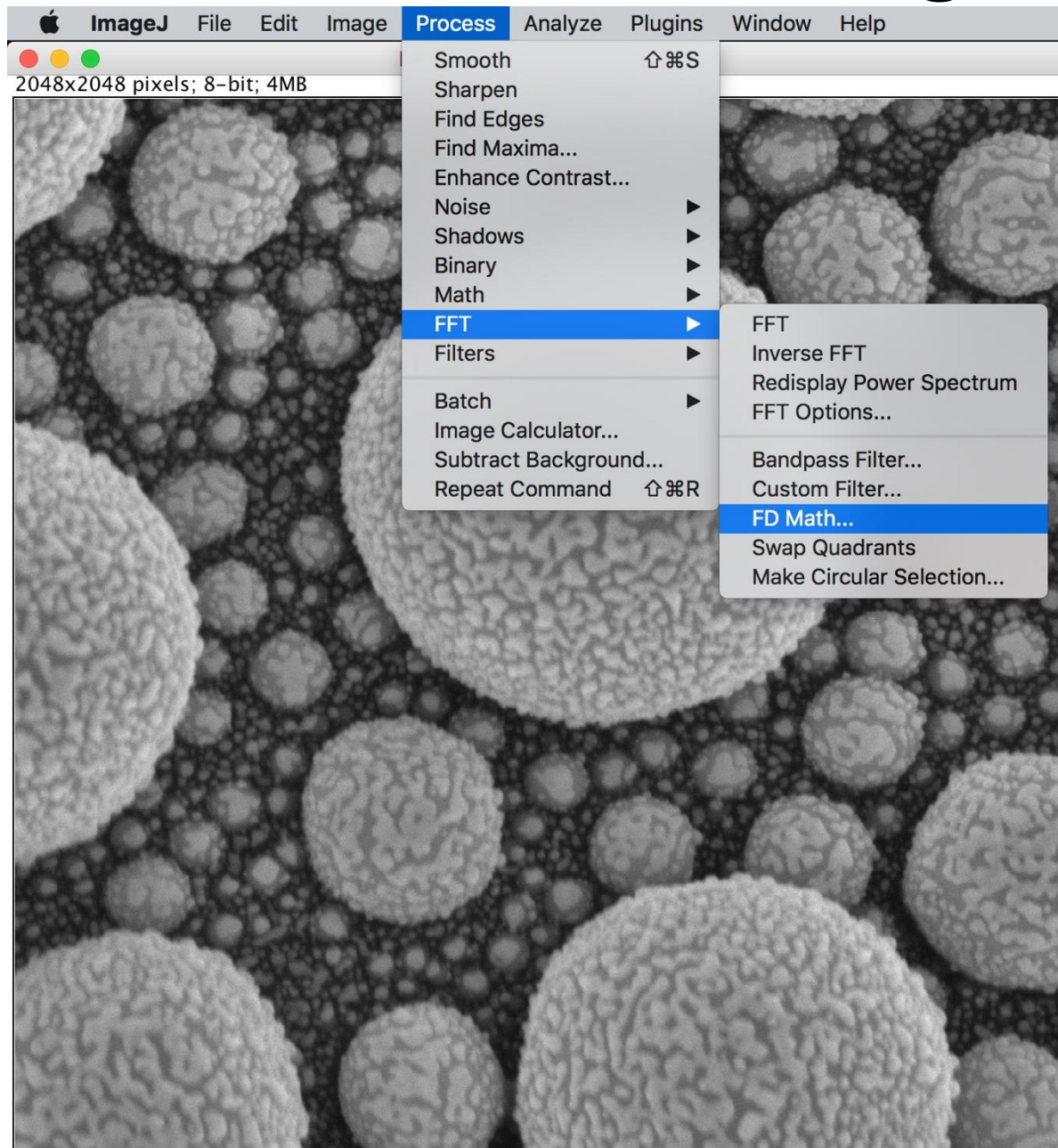
Autocorrelation in ImageJ

STEP 2: Open image in ImageJ

Canvas/Modules/Tutorials/Tute 1/balls1-2048x2048.tif

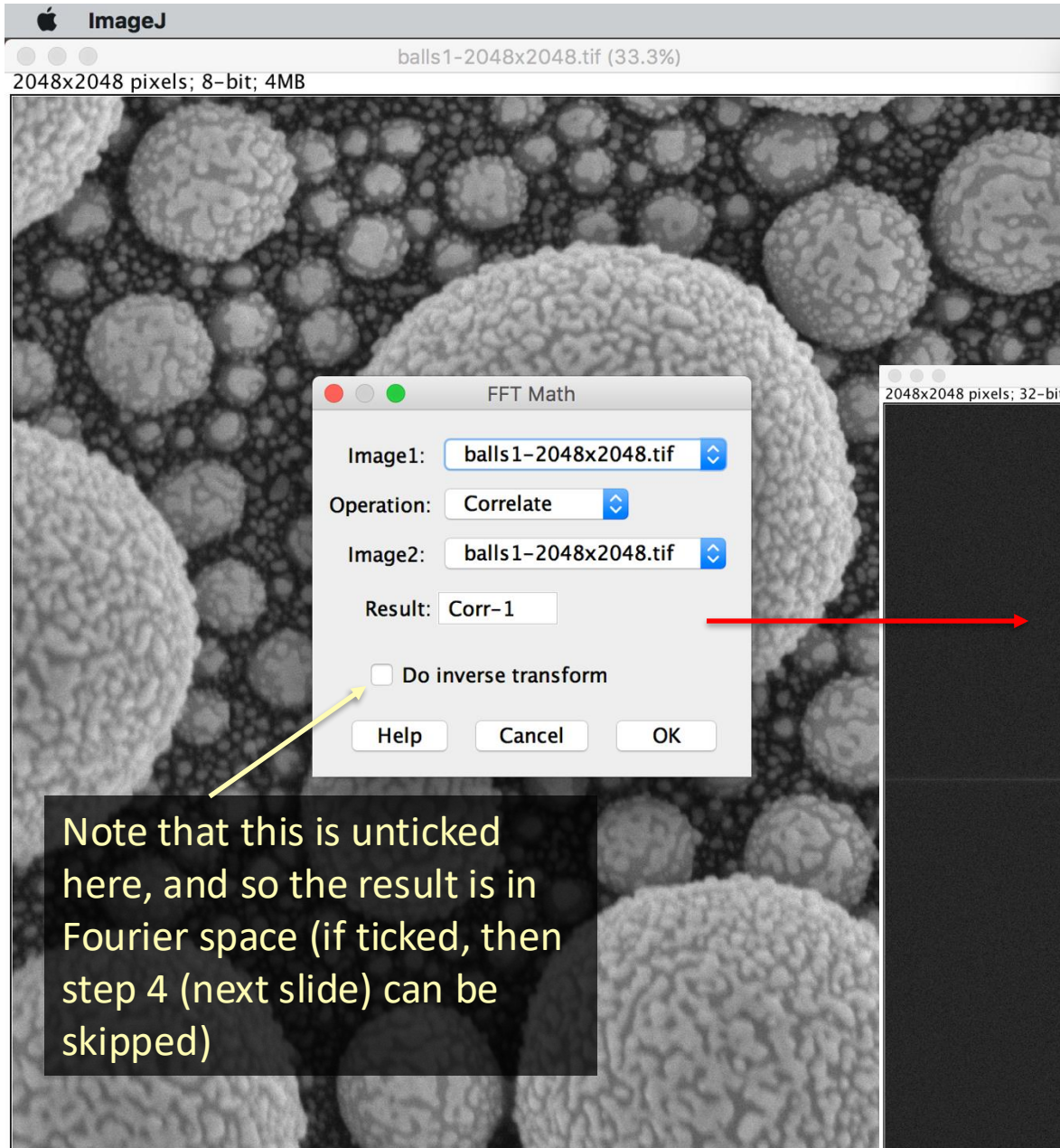


Autocorrelation in ImageJ

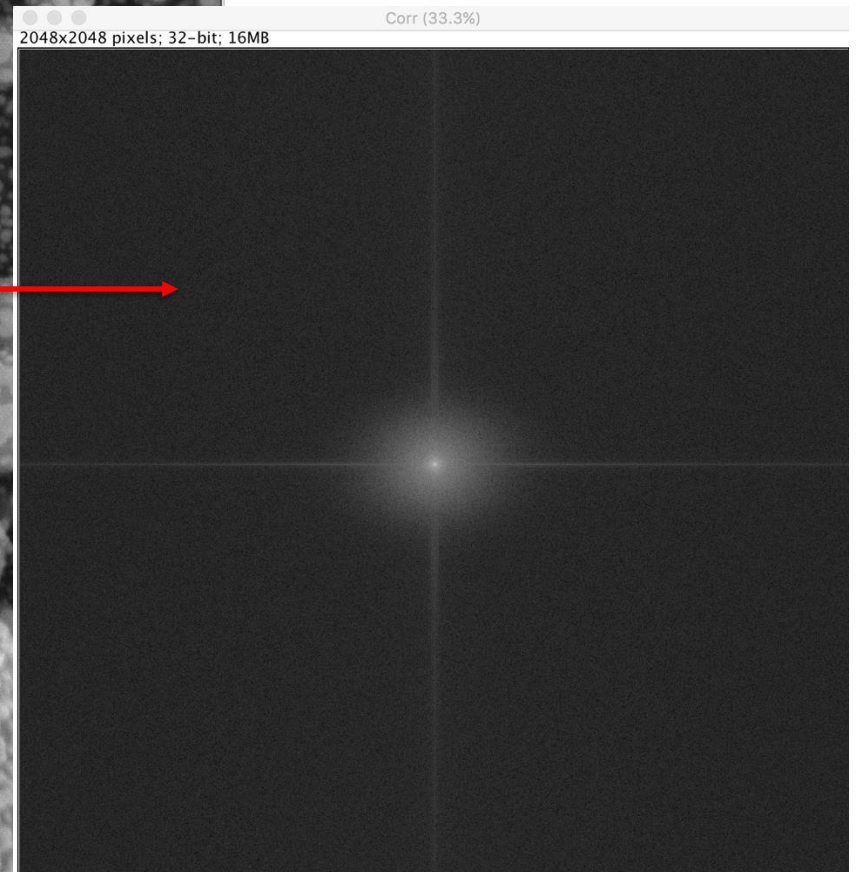


STEP 3: Perform autocorrelation in ImageJ using [FD Mat]...

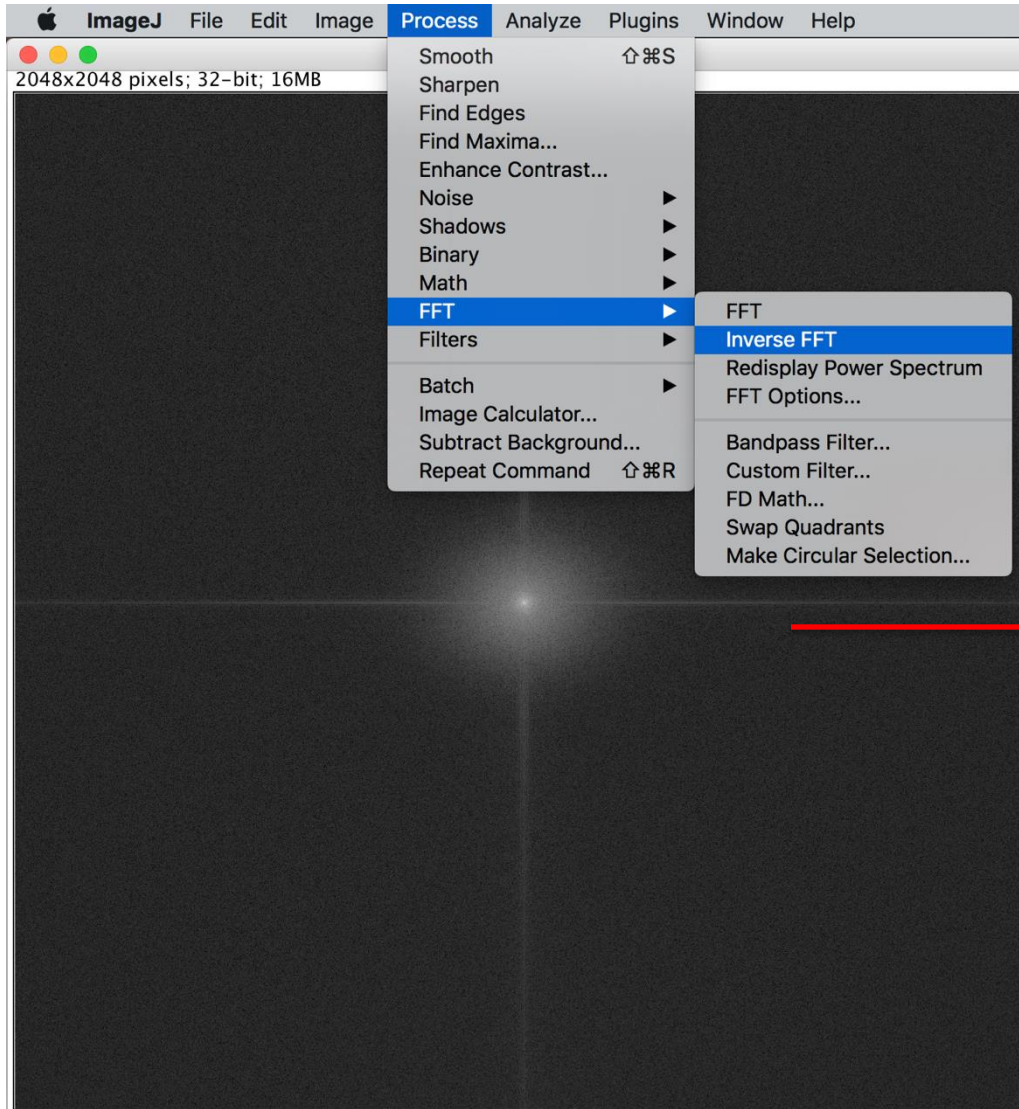
Autocorrelation in ImageJ



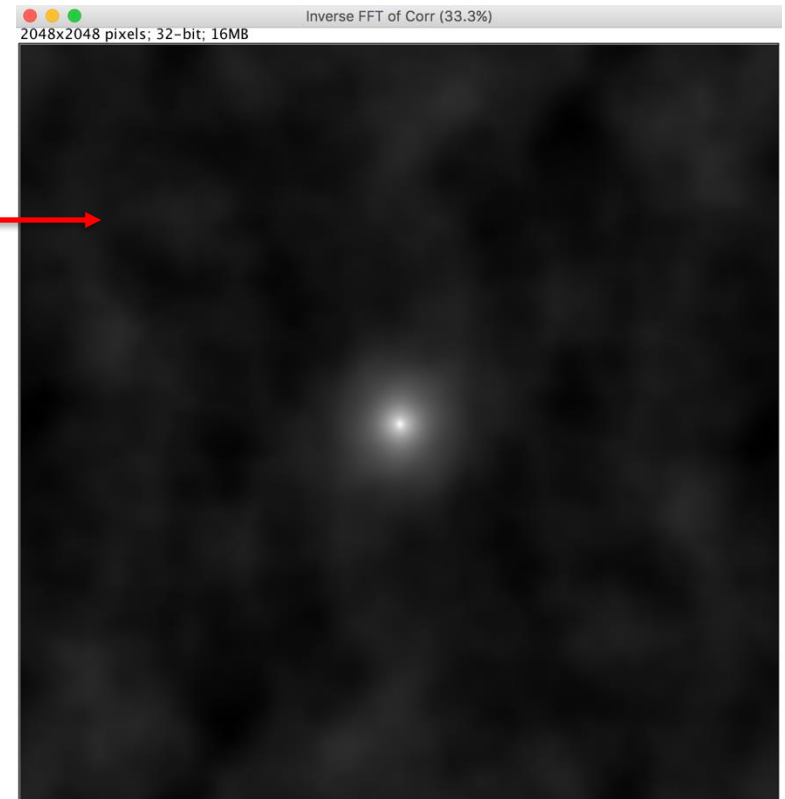
STEP 3: ImageJ – FD Math...



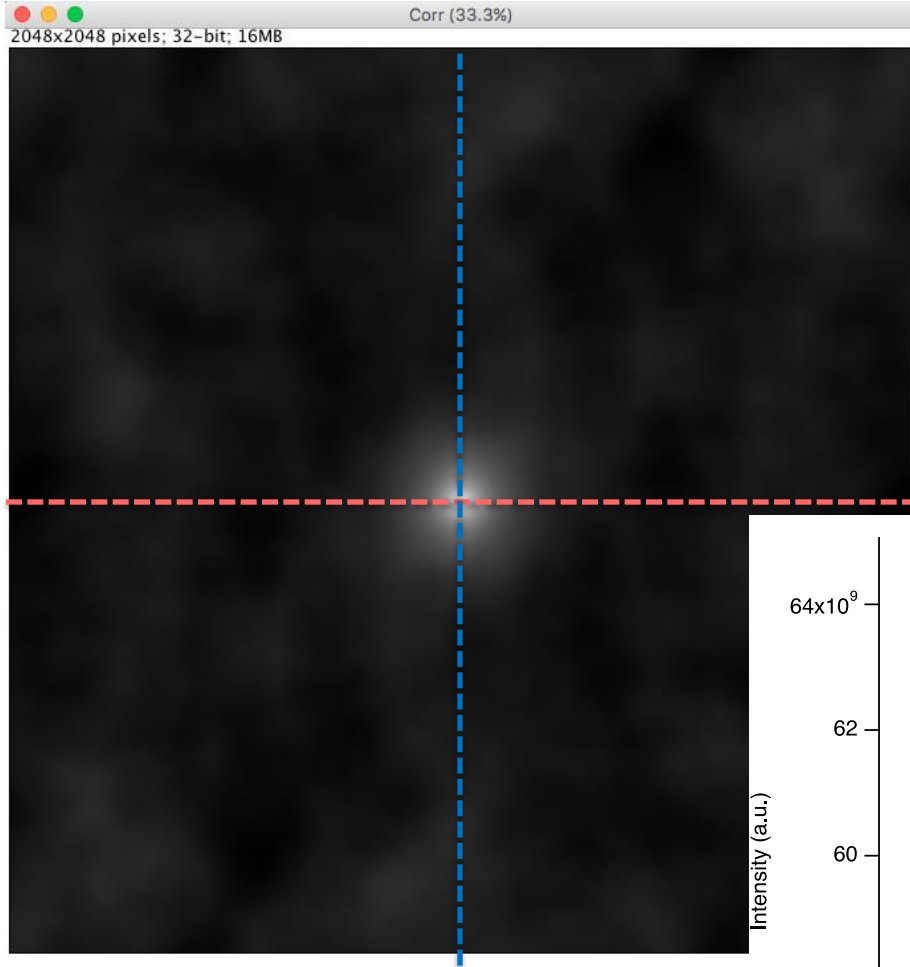
Autocorrelation in ImageJ



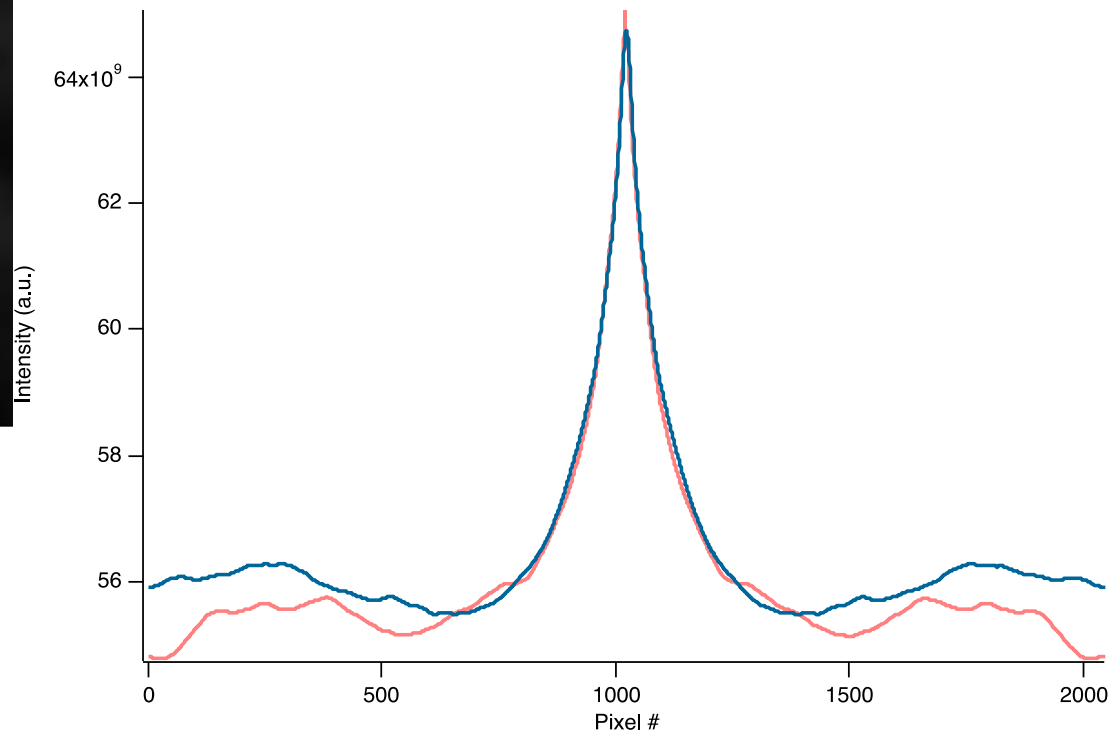
STEP 4: ImageJ – IFFT



Autocorrelation in ImageJ

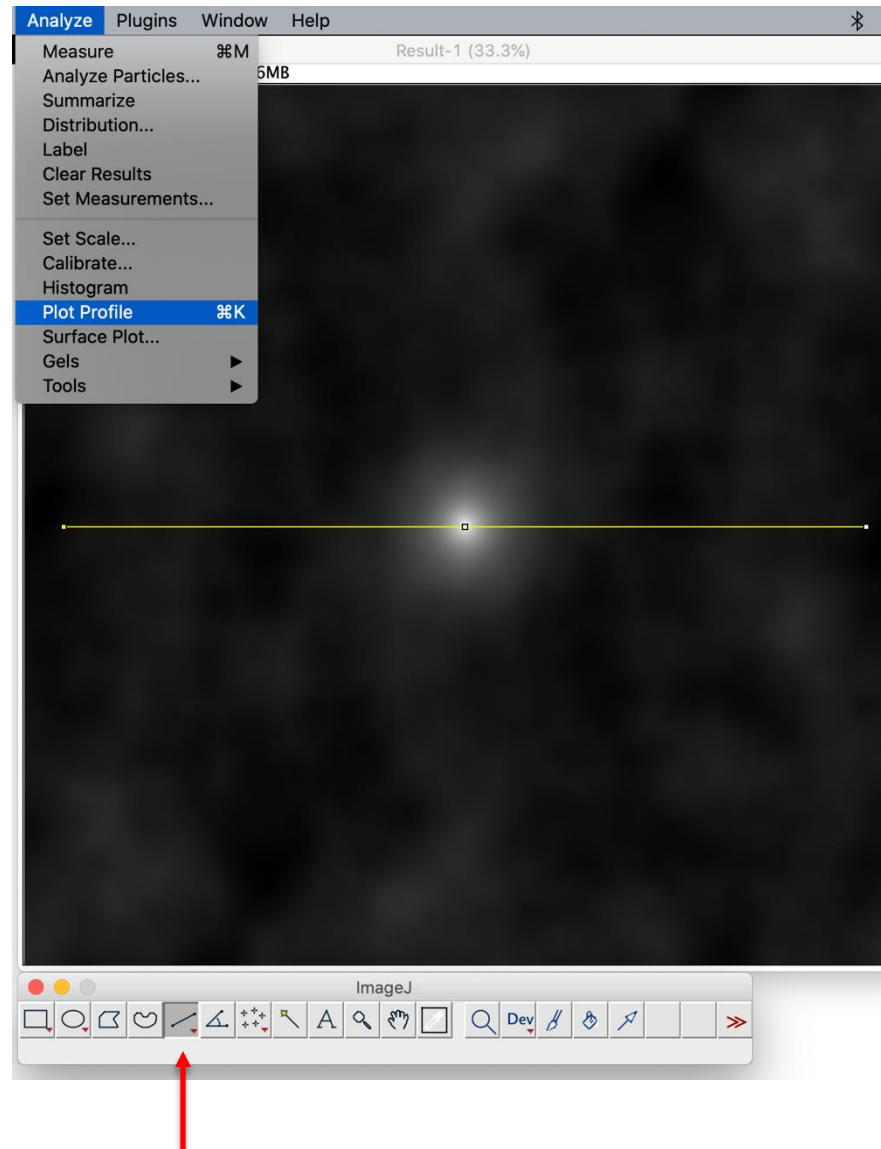


STEP 5: Plot horizontal & vertical slices through CCF
(see next slide)



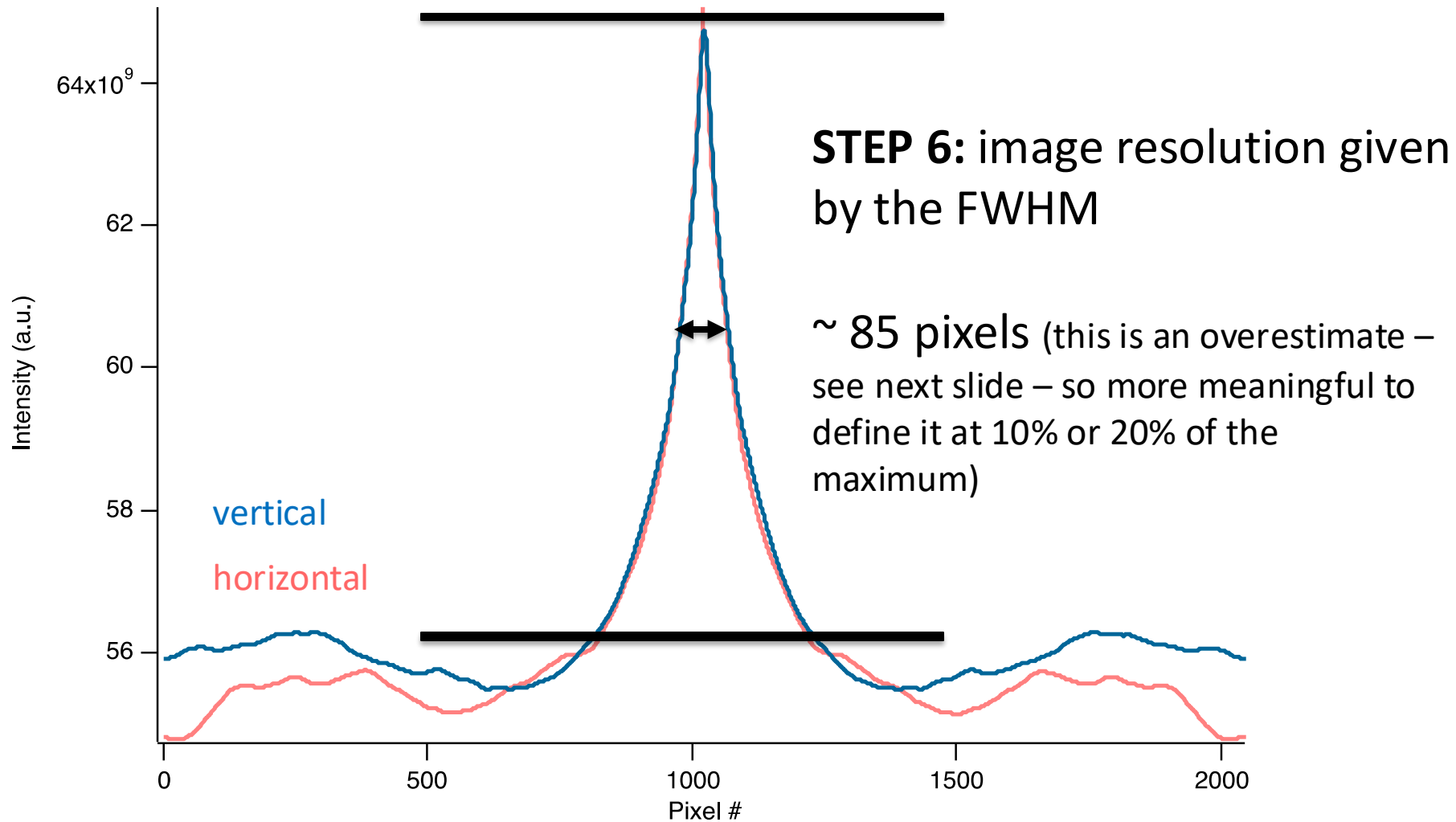
How to get a line profile in ImageJ

2) use this command
to generate a profile

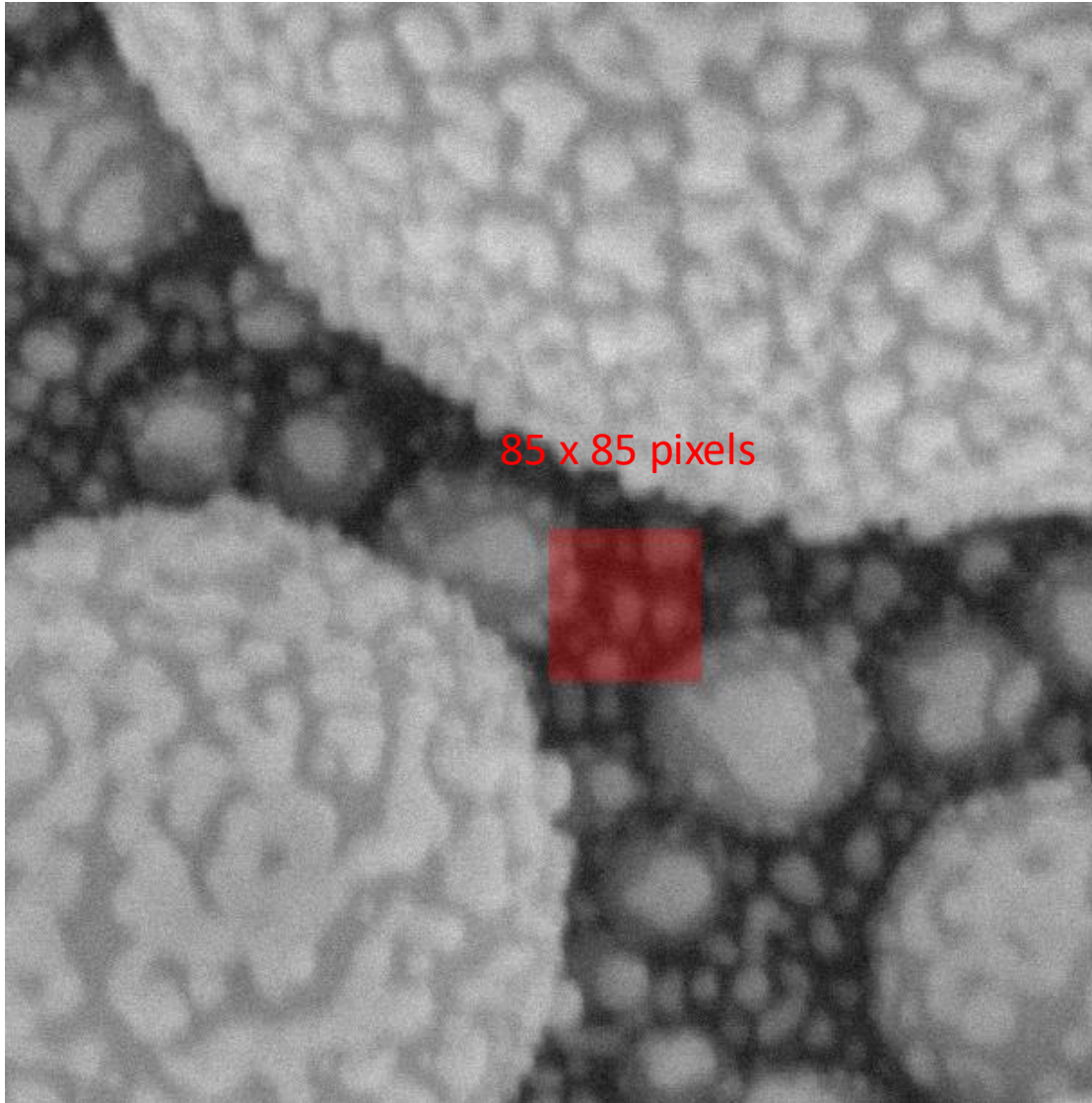


1) use this tool to define a line profile

Autocorrelation in ImageJ



Autocorrelation in ImageJ



1 pixel $\sim$ 0.8315 nm

85 pixels $\sim$ 71 nm

Bottom line: The method doesn't yield true resolution but it is more robust & reproducible than the other methods we looked at. Hence, useful for monitoring and comparing the imaging performance of SEMs over time

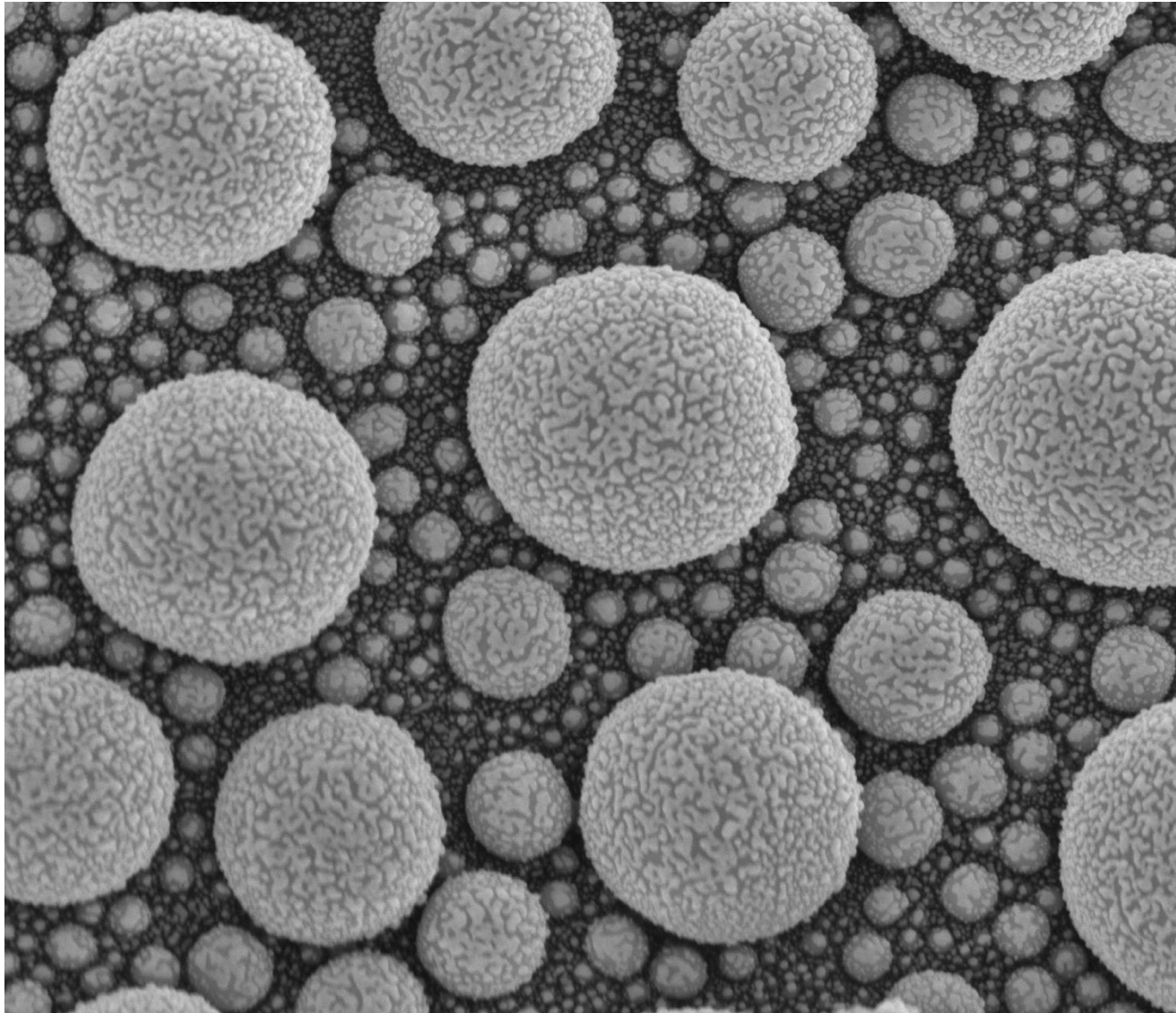
Things to be aware of when measuring SEM image resolution

can you explain the reason
behind each of these?

- Resolution is sample-dependent
- It's important to ensure that the magnification is sufficiently high so that the distance corresponding to 1 pixel is smaller than the image resolution
- The image signal must not be clipped (black or white)
- The image signal-to-noise ratio must be high enough to resolve the features of interest
- The image must not be astigmatic
- Resolution should be measured in both x & y directions

Using the cross-correlation function (CCF) to measure image drift between successive image frames

Cross-correlation function (CCF)



STEP 1

Image of sample

FOV = 2980 nm

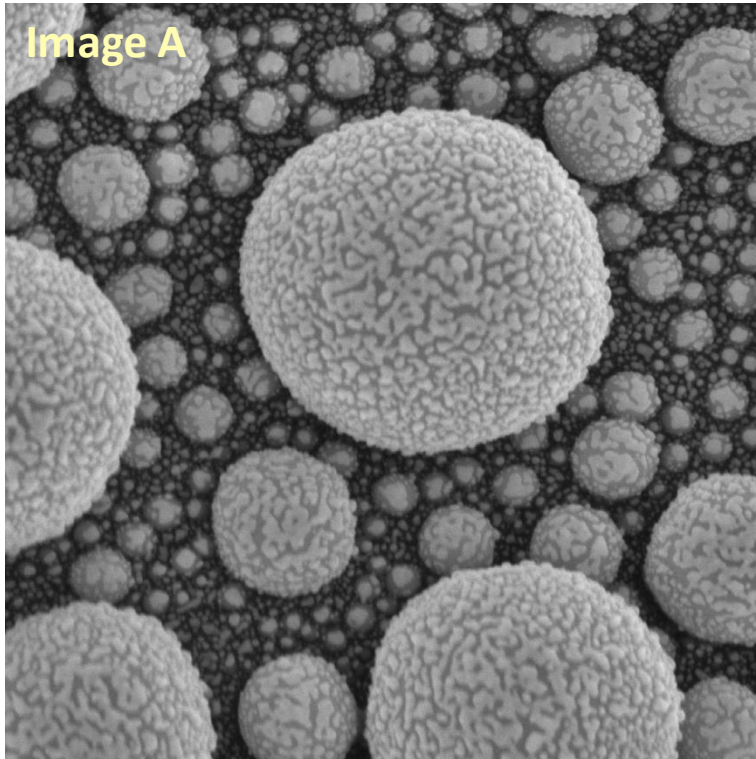
FOV = 3584 pixels

ie: 1 pixel $\sim$ 0.8315 nm

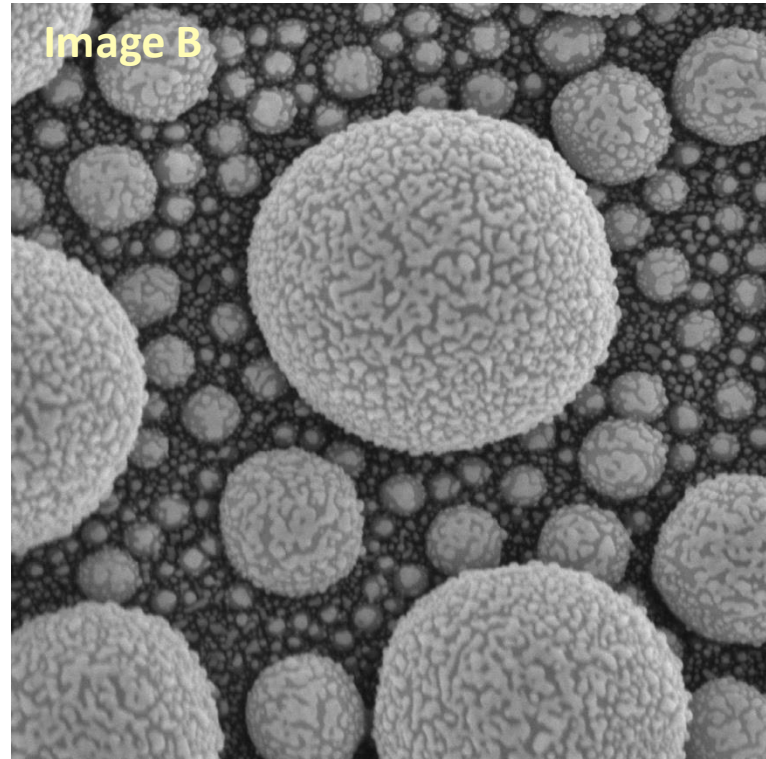
Cross-correlation function (CCF)

For this demonstration, 2 extracts of the image were made, and offset by 4 pixels.

STEP 2: Load 2 extracts of the image into ImageJ



2048 x 2048 pixels



2048 x 2048 pixels,
offset horizontally by 4 pixels

Cross-correlation function (CCF)

STEP 3: ImageJ – FD Math...

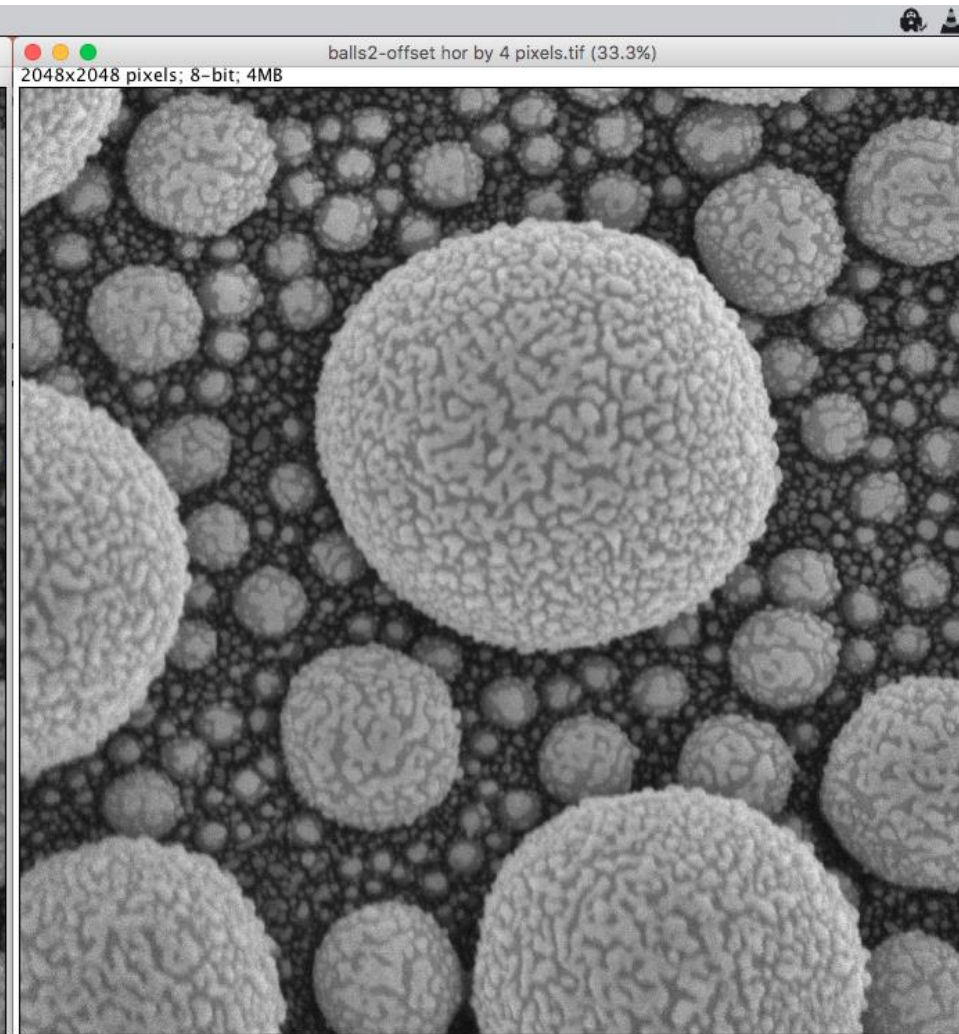
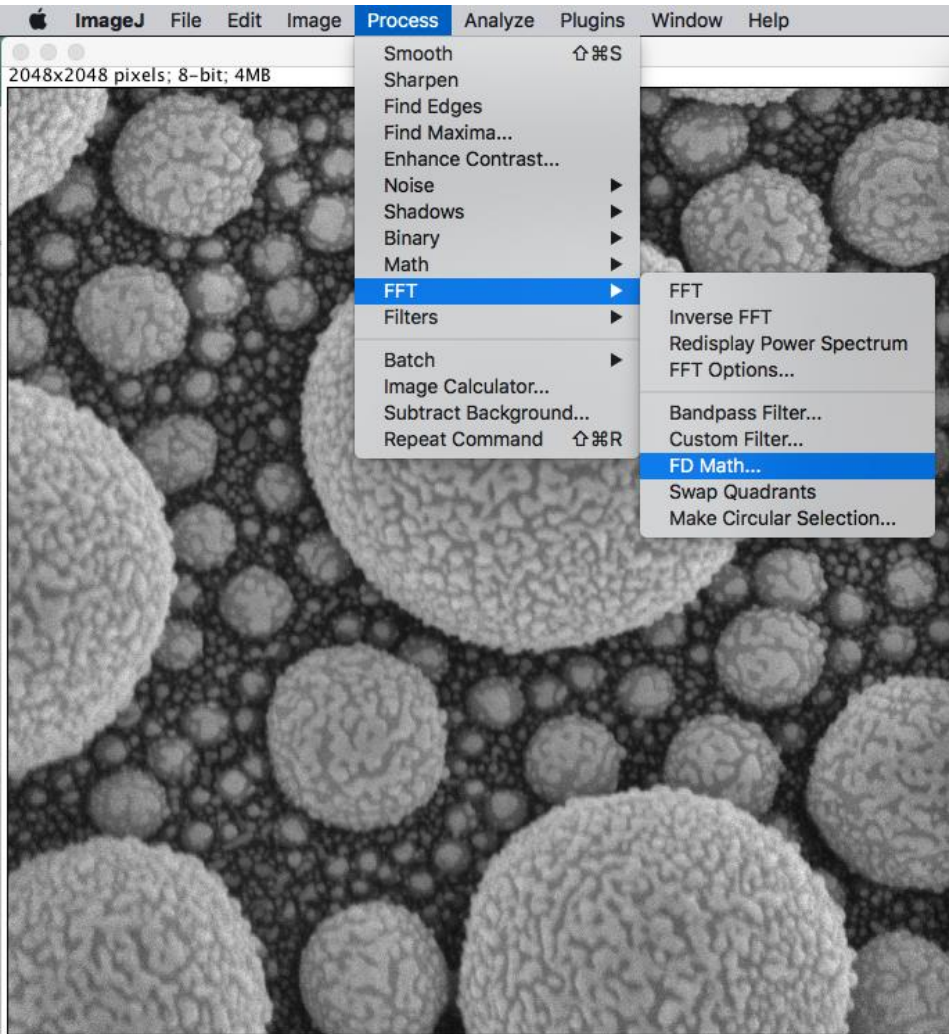


Image A

Canvas/Modules/Tutorials/Tute 1 /balls1-2048x2048.tif

Image B

Canvas/Modules/Tutorials/Tute 1/balls2-offset hor by 4 pixels.tif

Cross-correlation function (CCF)

STEP 4: ImageJ – CCF

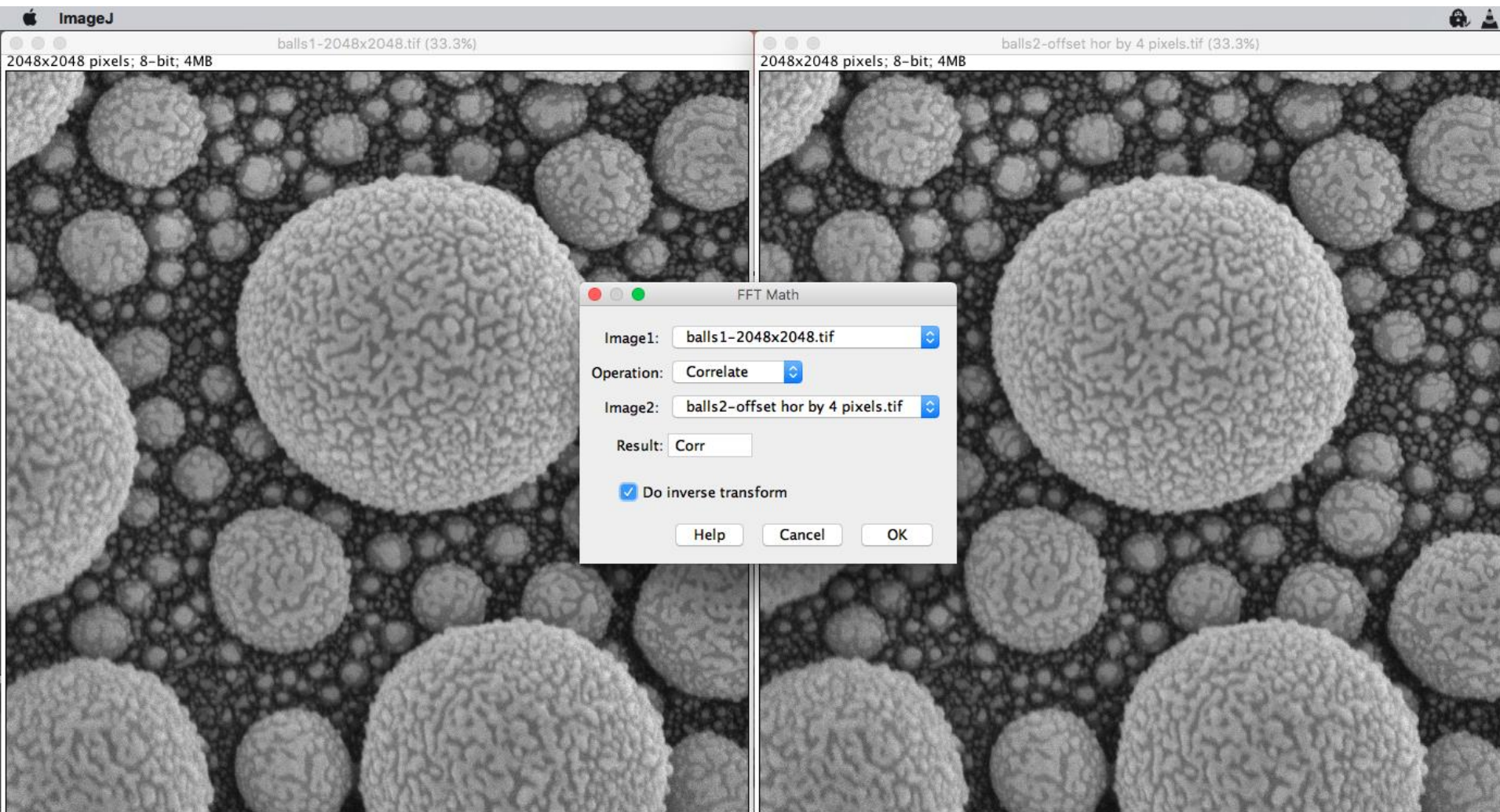
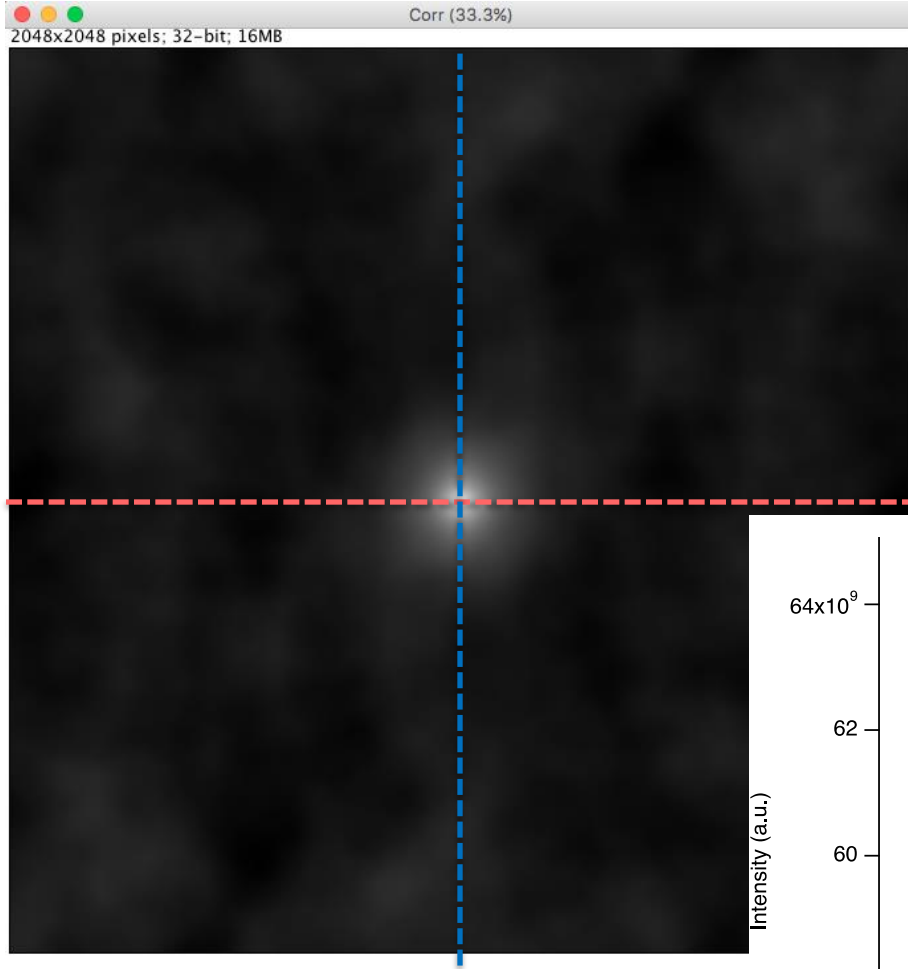


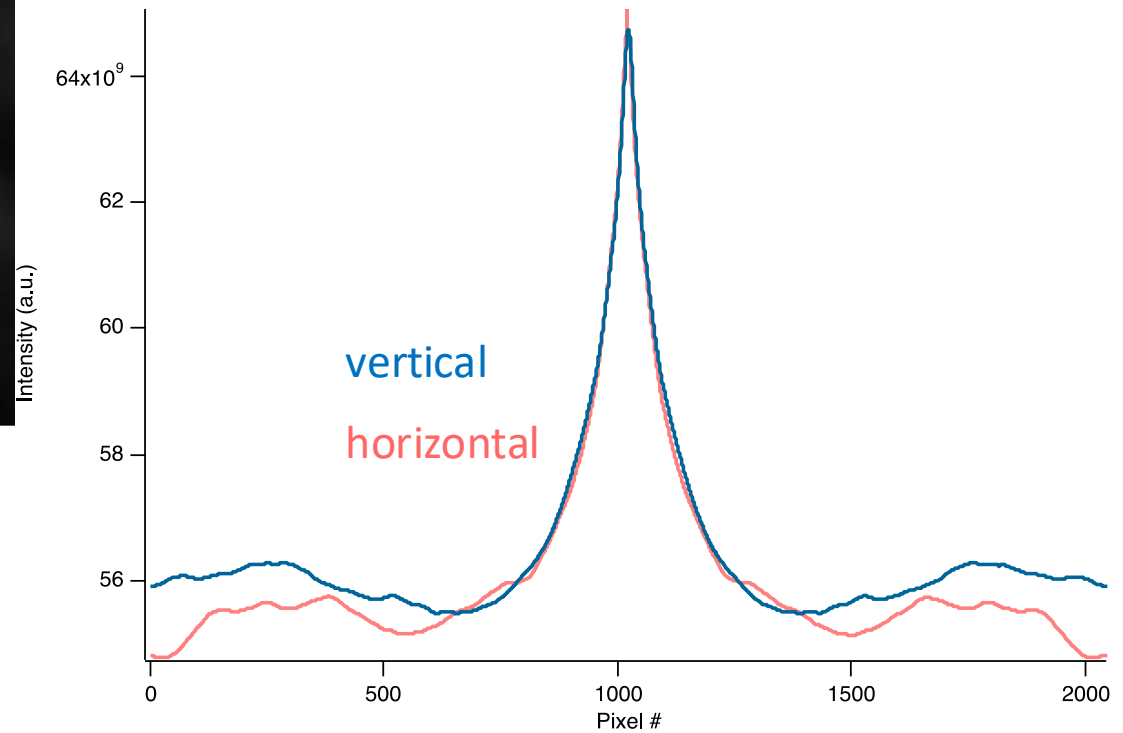
Image A

Image B

Cross-correlation function (CCF)

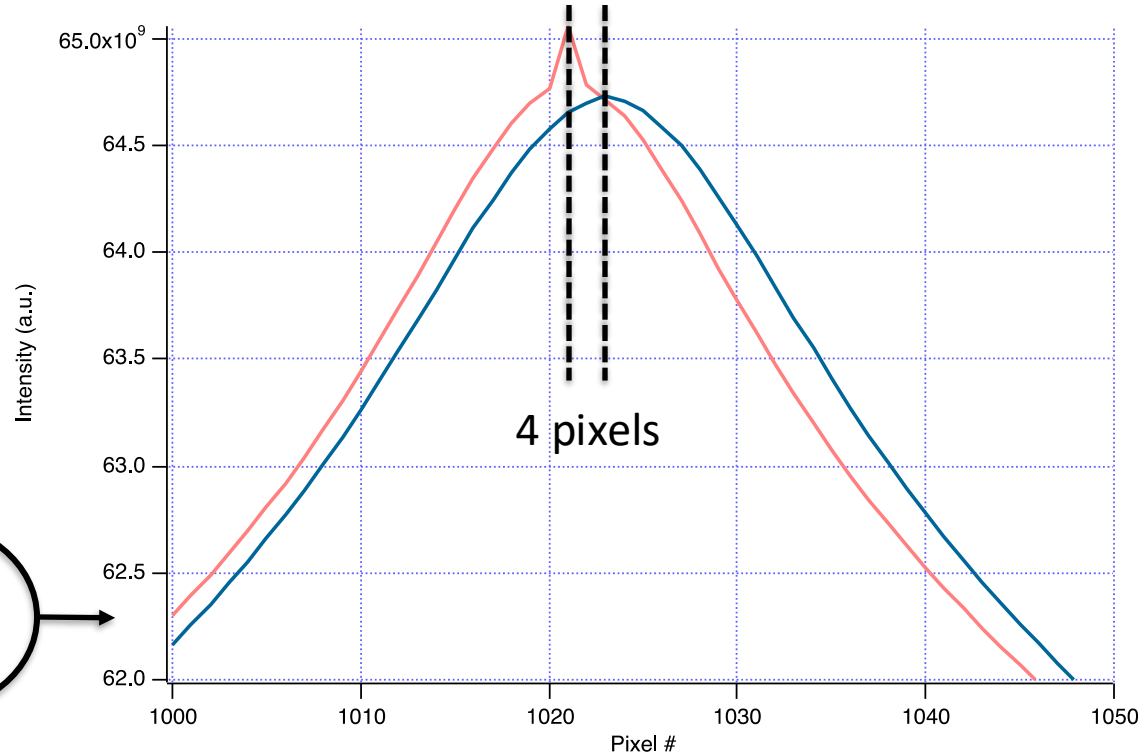
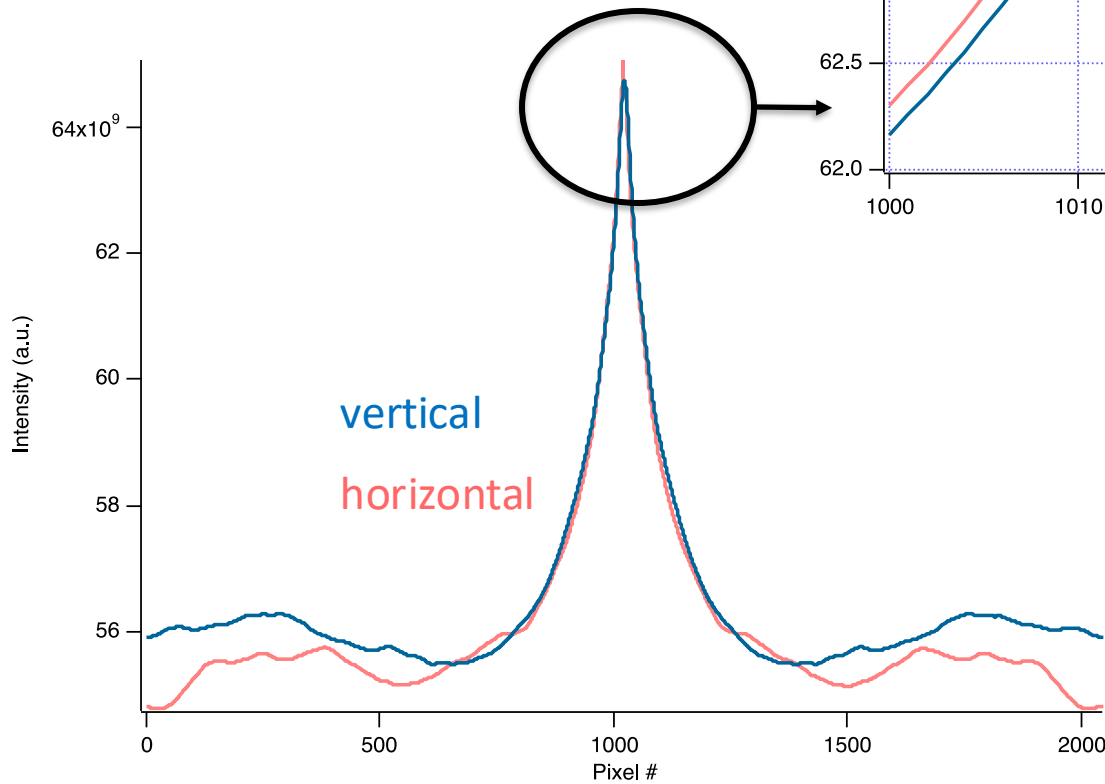


Plot horizontal & vertical
slices through CCF



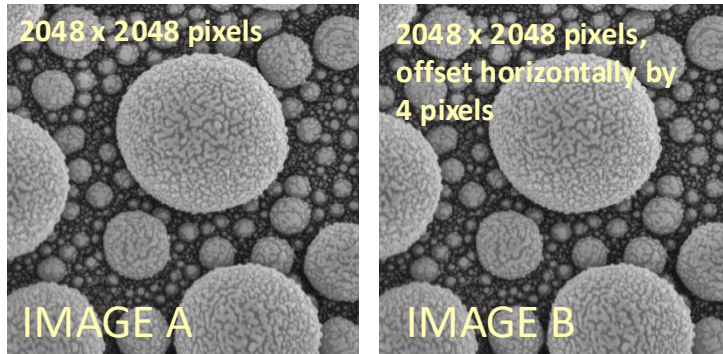
Cross-correlation function (CCF)

Zoom in & compare
peak positions

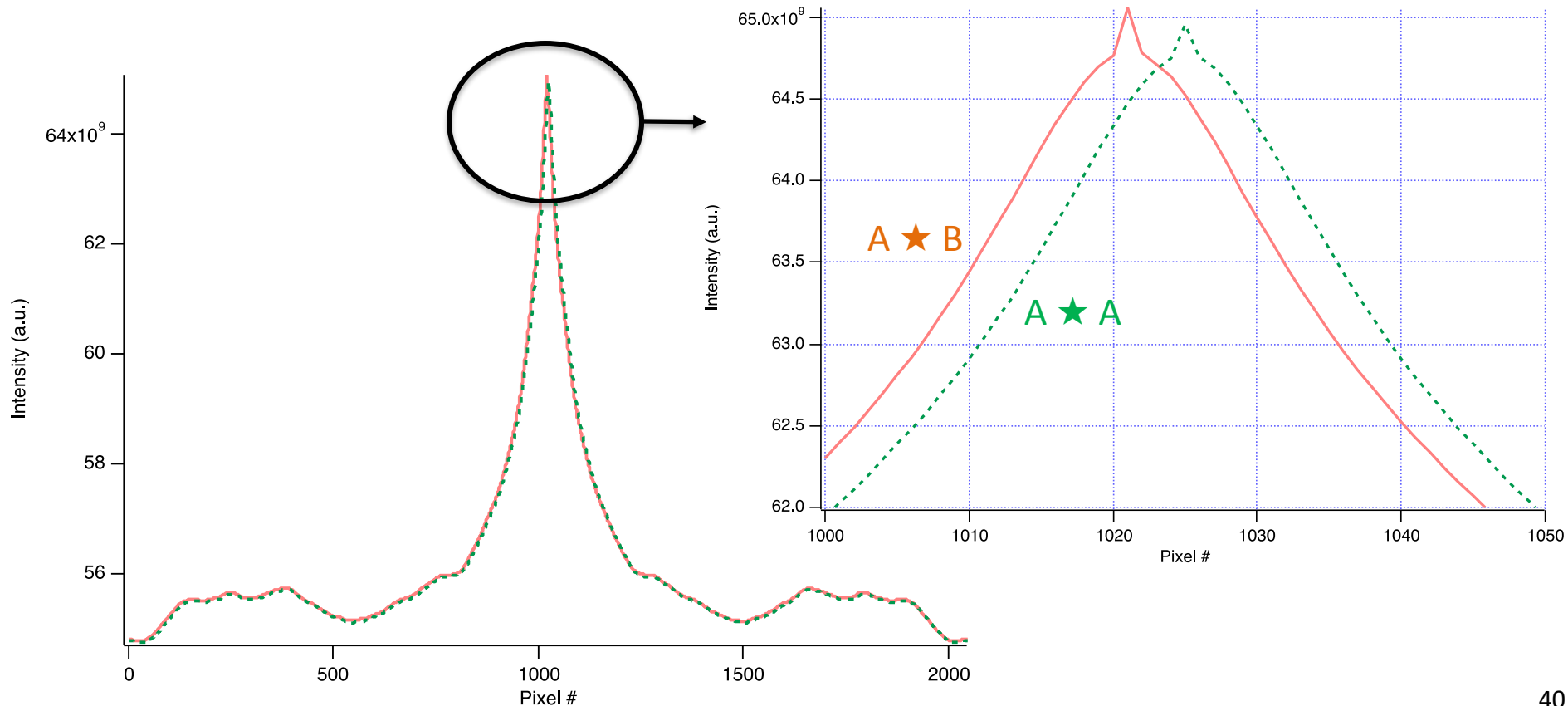


recall that we offset the 2
images (A & B) horizontally by
4 pixels

Cross-correlation function (CCF)



- we can also compare the horizontal cross-correlation function ($A \star B$) to the autocorrelation function ($A \star A$)
- again, the peaks are offset by 4 pixels (as expected)



Reading material

The following paper is in Canvas:

Canvas/Modules/Tutorials/Tute 1/06. SEM Resolution.pdf

This paper was published here: Proc. SPIE 3998, Metrology, Inspection, and Process Control for Microlithography XIV, (2 June 2000); doi: 10.1117/12.386444

Metrics of resolution and performance for CD-SEMs

David C Joy^{a,b}, Yeong-Uk Ko^a, and Justin J Hwu^a

^aEM Facility, University of Tennessee, Knoxville, TN 37996

^bOak Ridge National Laboratory, Oak Ridge, TN 37831

This program is now known as "ImageJ" or "FIJI". It's implemented in Java and runs on both Macs and Windows machines. It's freeware & can be downloaded from the web

ABSTRACT

The performance of scanning electron beam instruments such as CD-SEMs can be defined in terms of parameters such as the beam probe size, the spatial resolution, and the signal to noise ratio of the image. A knowledge of these quantities is important in verifying the fact that an instrument meets its specification, and subsequently for tracking and optimizing its performance during use. Analytical methods based on the power spectrum (2-D Fourier transform analysis) of images are now beginning to be used for these purposes but care must be exercised to ensure reliable and meaningful results. Two new methods are suggested which can offer more detailed information about the microscope performance while avoiding the pitfalls of the simpler technique. Code implementing these tests, written as a plug-in macro for the well known NIH Image program, is available on-line.

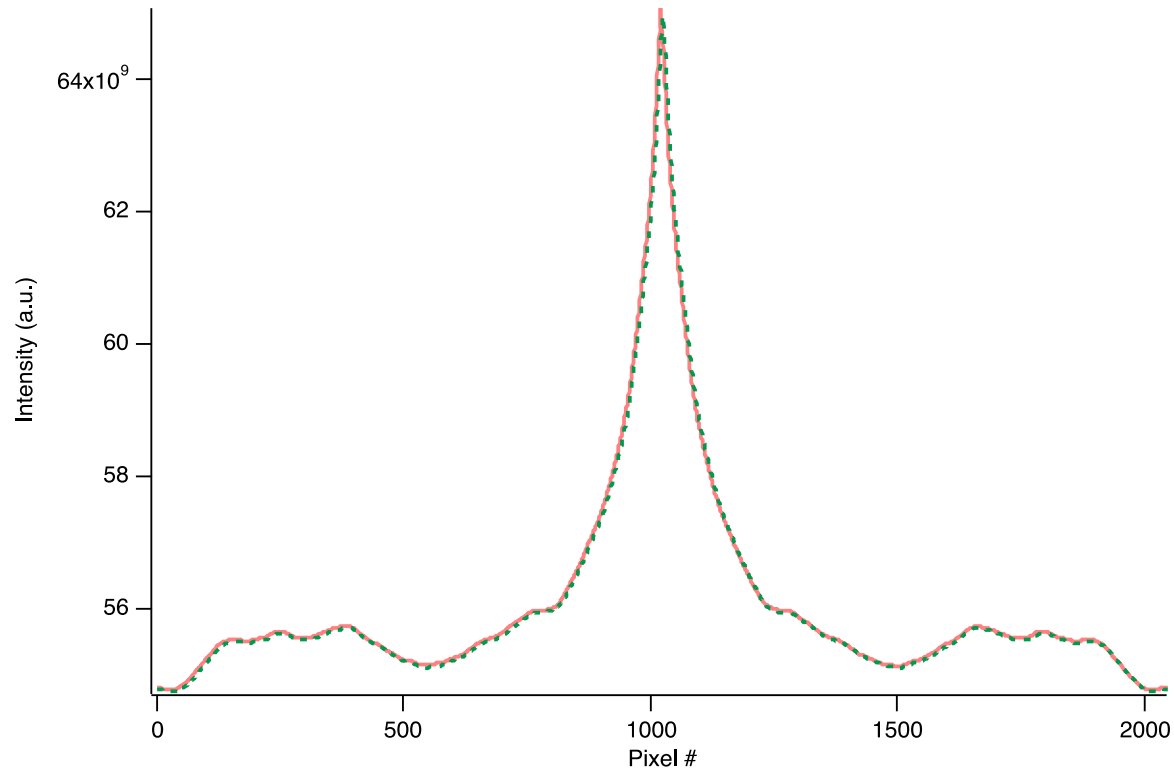
Keywords: scanning electron microscope, resolution, signal to noise, probe size, Fourier transform

1. INTRODUCTION

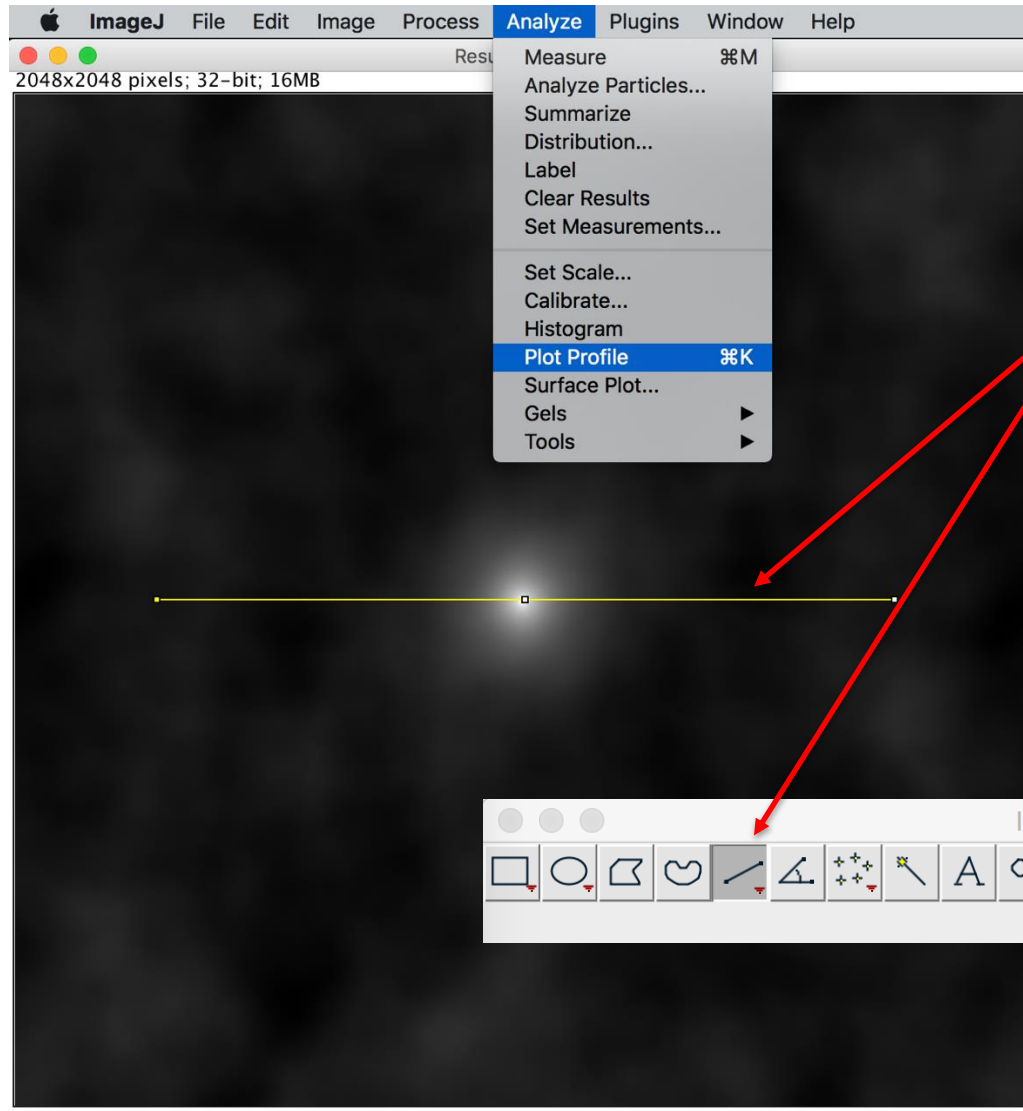
Scanning electron microscopes designed for the critical dimension metrology (CD-SEMs) must achieve a resolution approaching 1nm for the current technology nodes¹. Verifying that the tool is reaching the required level of performance when it is delivered, and subsequently monitoring its performance during use, requires that tests be available which can quantify the parameters such as probe size, resolution, the signal to noise ratio of the image under standard operating conditions etc., which define the performance level. In addition many electron beam tools are now used in a semi-automatic fashion with little or no operator supervision. There is thus a need for techniques which can monitor the performance of the tool to ensure that it remains within acceptable limits. Ideally such tests should be capable of being performed

APPENDIX A

How to get these curves from Image J?

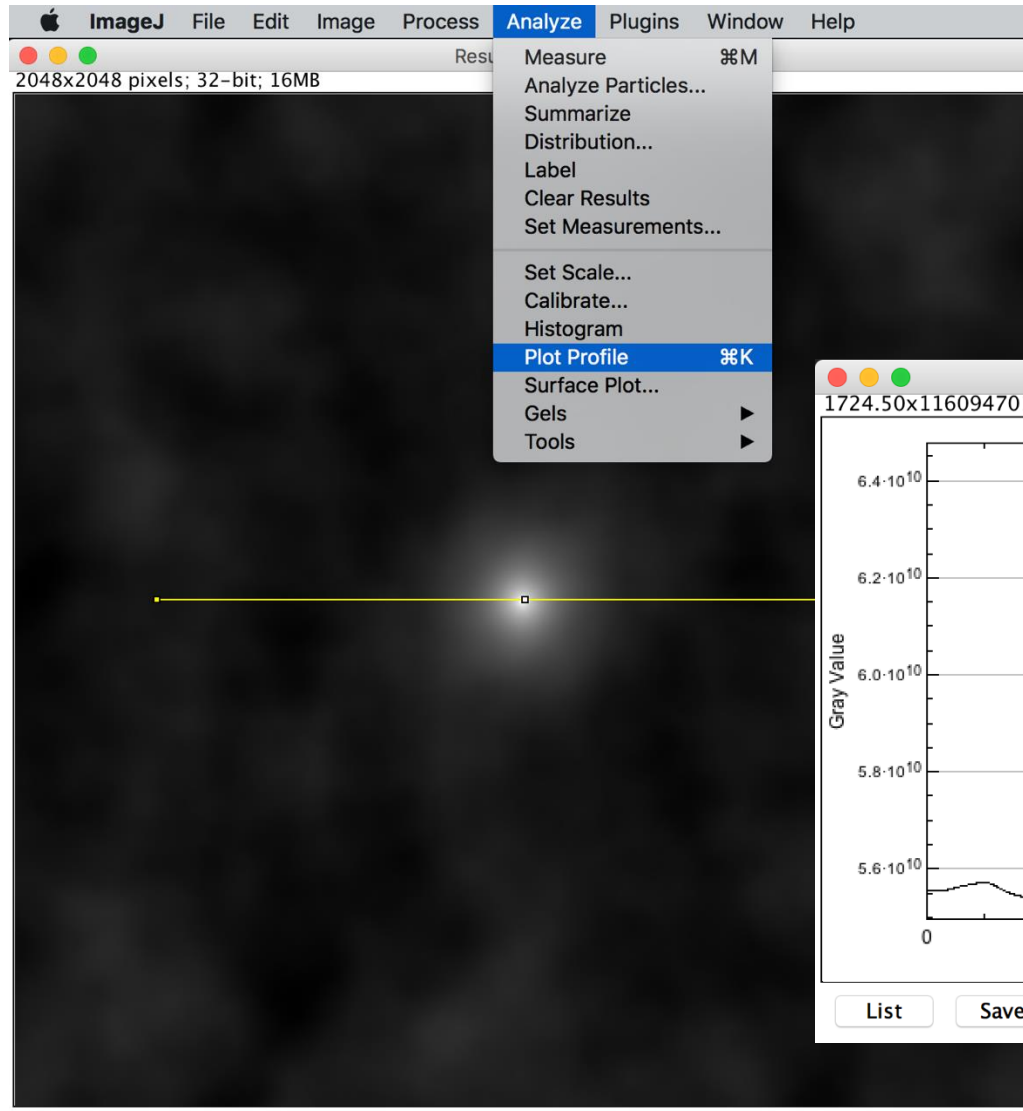


How to get these curves from Image J?

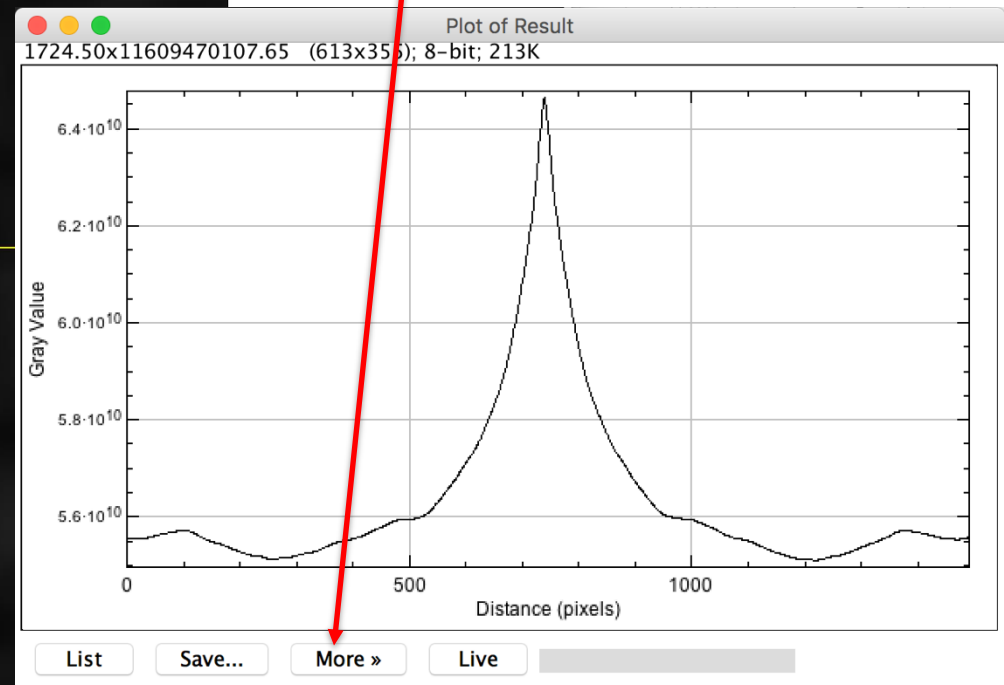


1. Select a "region" using the line tool
2. Click on Plot Profile...

How to get these curves from Image J?



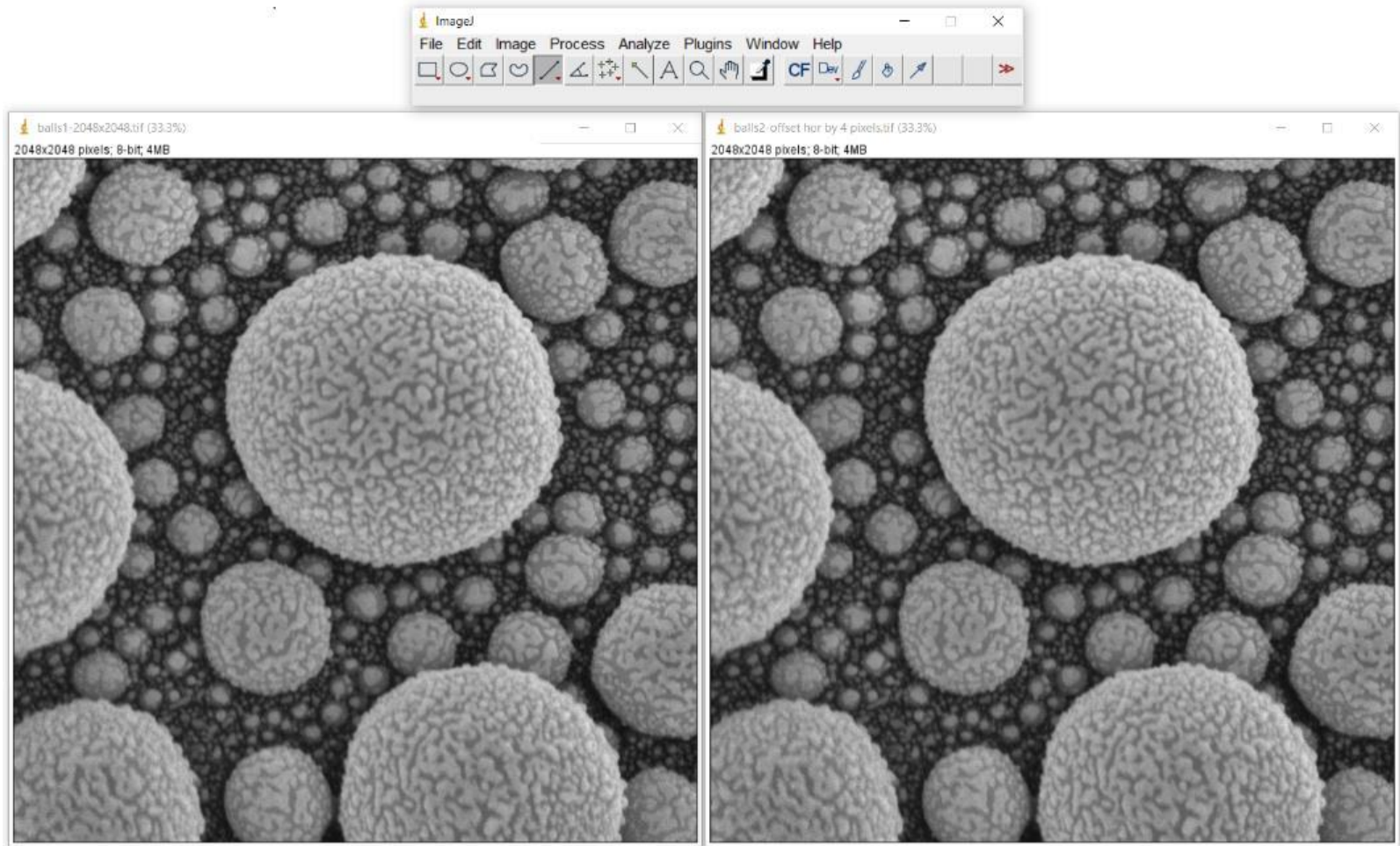
3. Analyze the plot in ImageJ or copy/paste the data into Excel (or whatever data analysis package you use)...



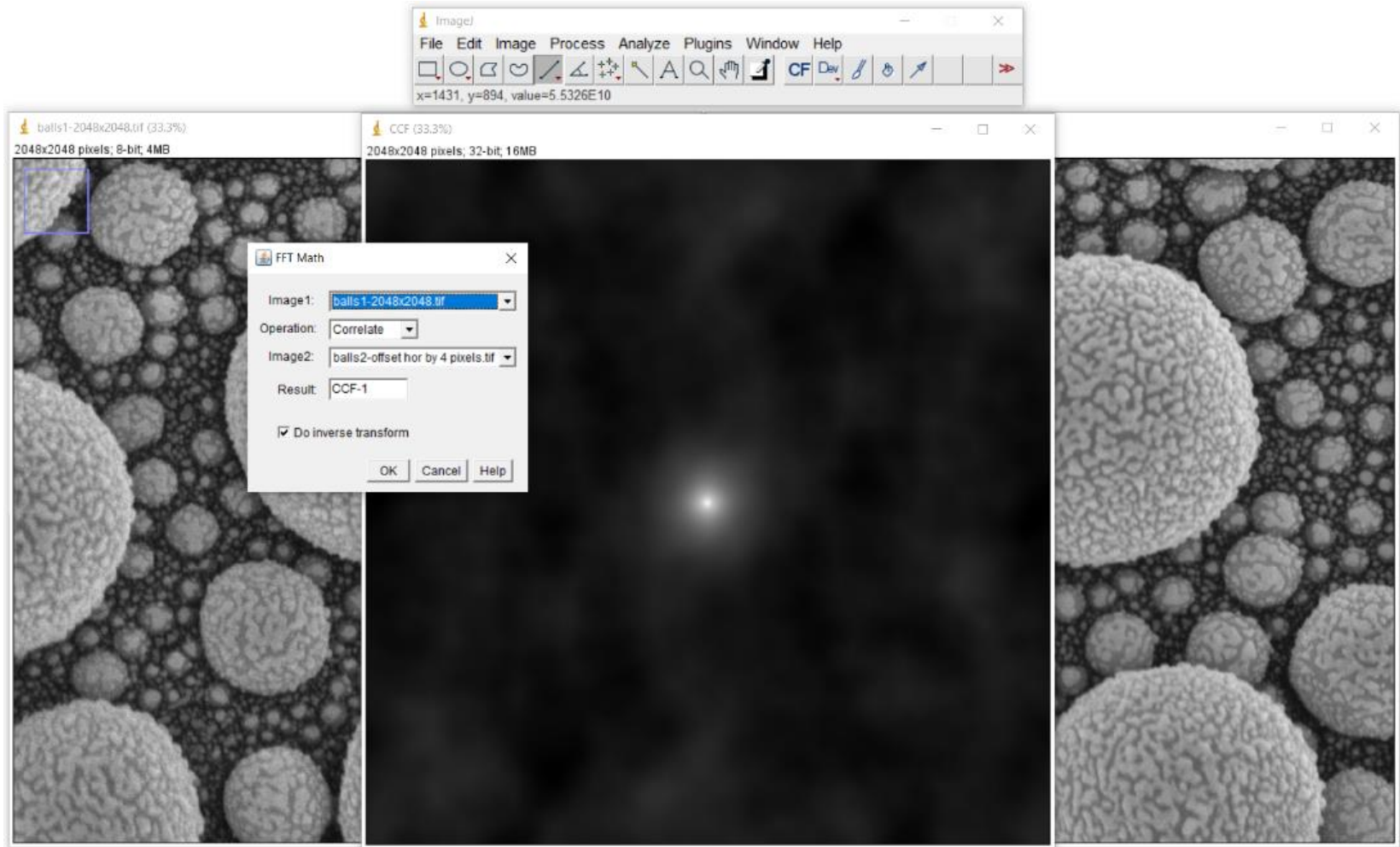
APPENDIX B

- **Perfect line profiles in ImageJ**
- **Cross-correlation function analysis macro in ImageJ**
- **Plotting 2 lines in ImageJ**

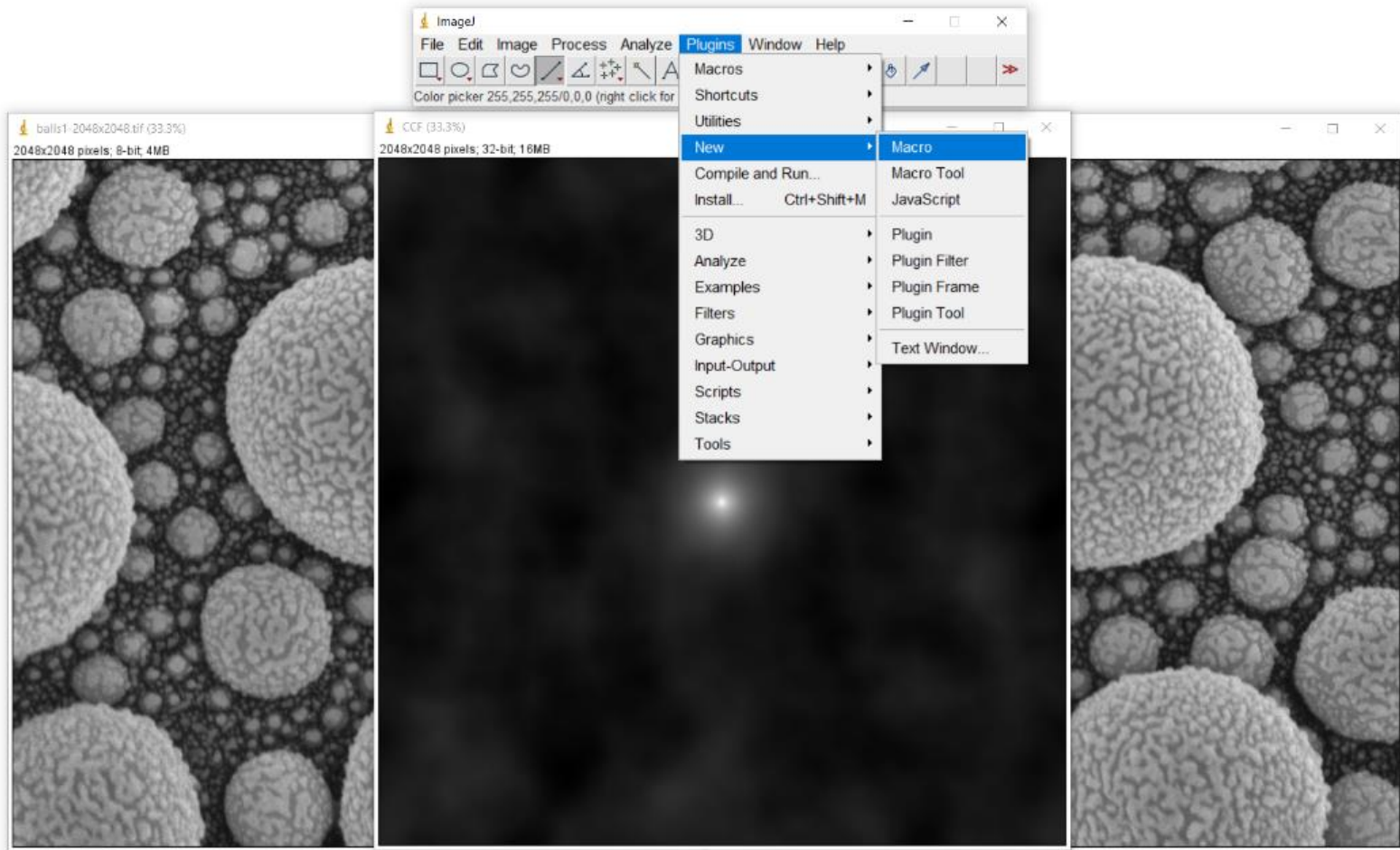
Load in both images at the same time, put them side-by-side



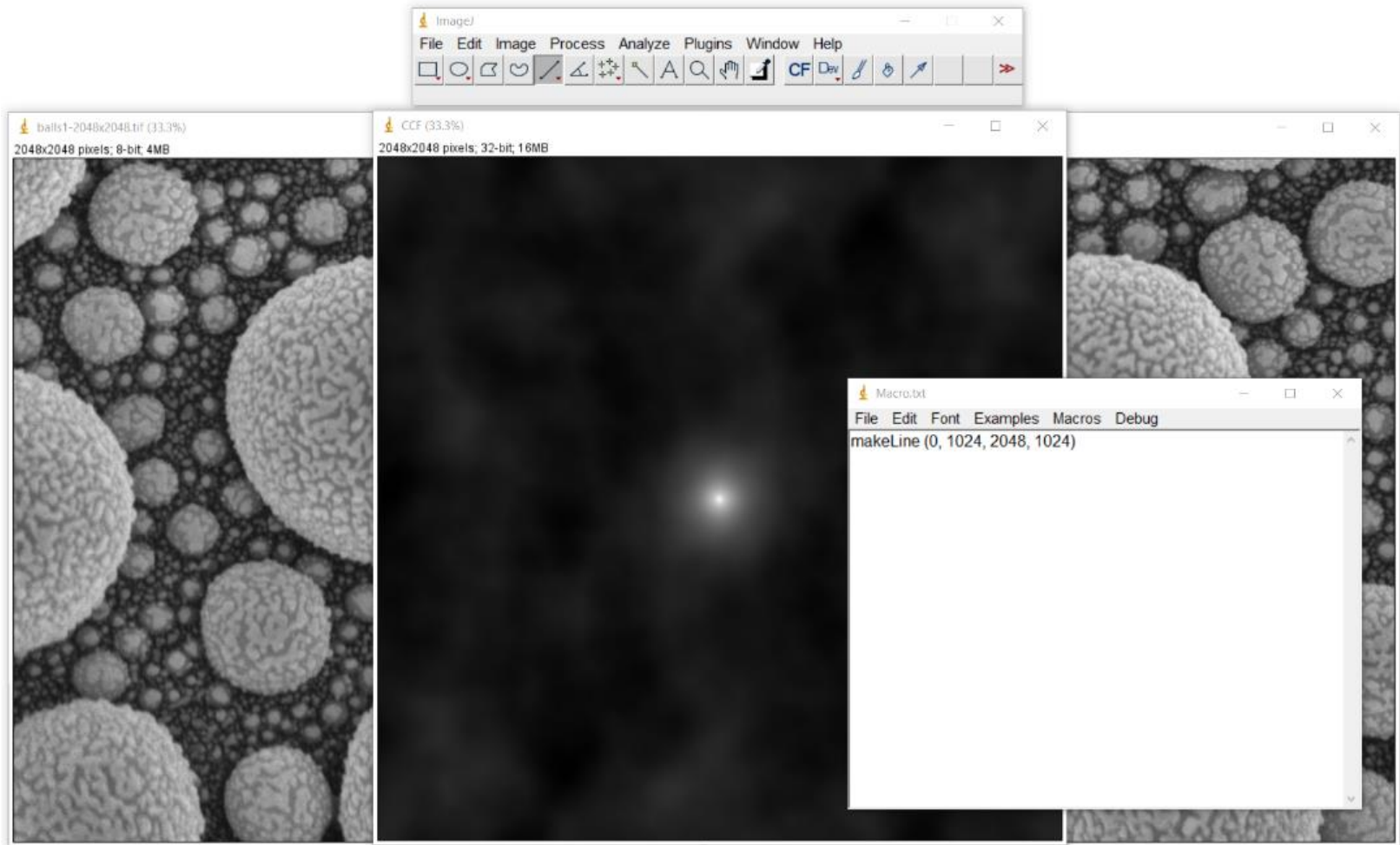
Use FD math to obtain the cross correlation function (CCF)



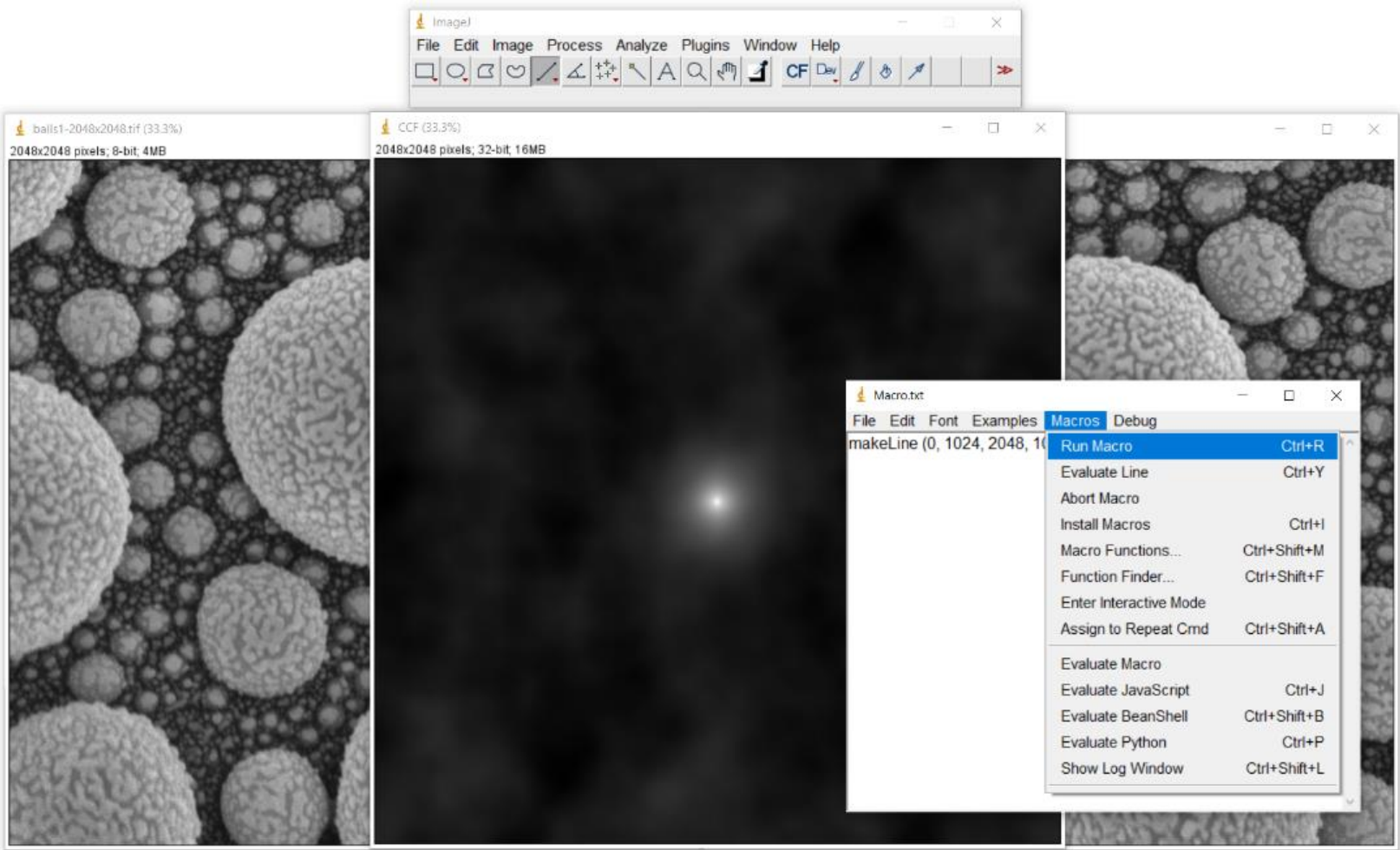
Go to Plugins > New > Macro



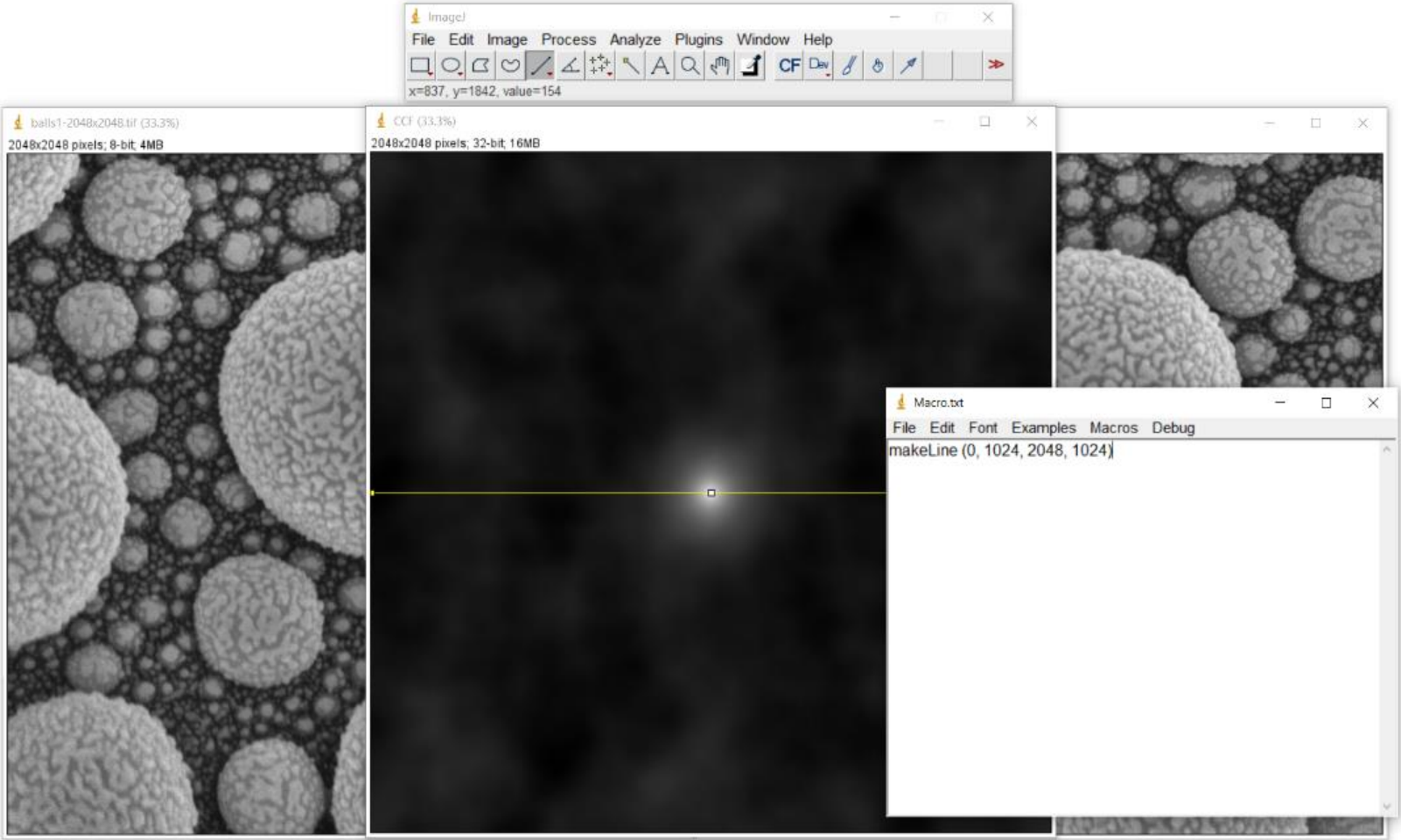
Type “makeLine (0, 1024, 2048, 1024)” these are the starting and ending coordinates for the line



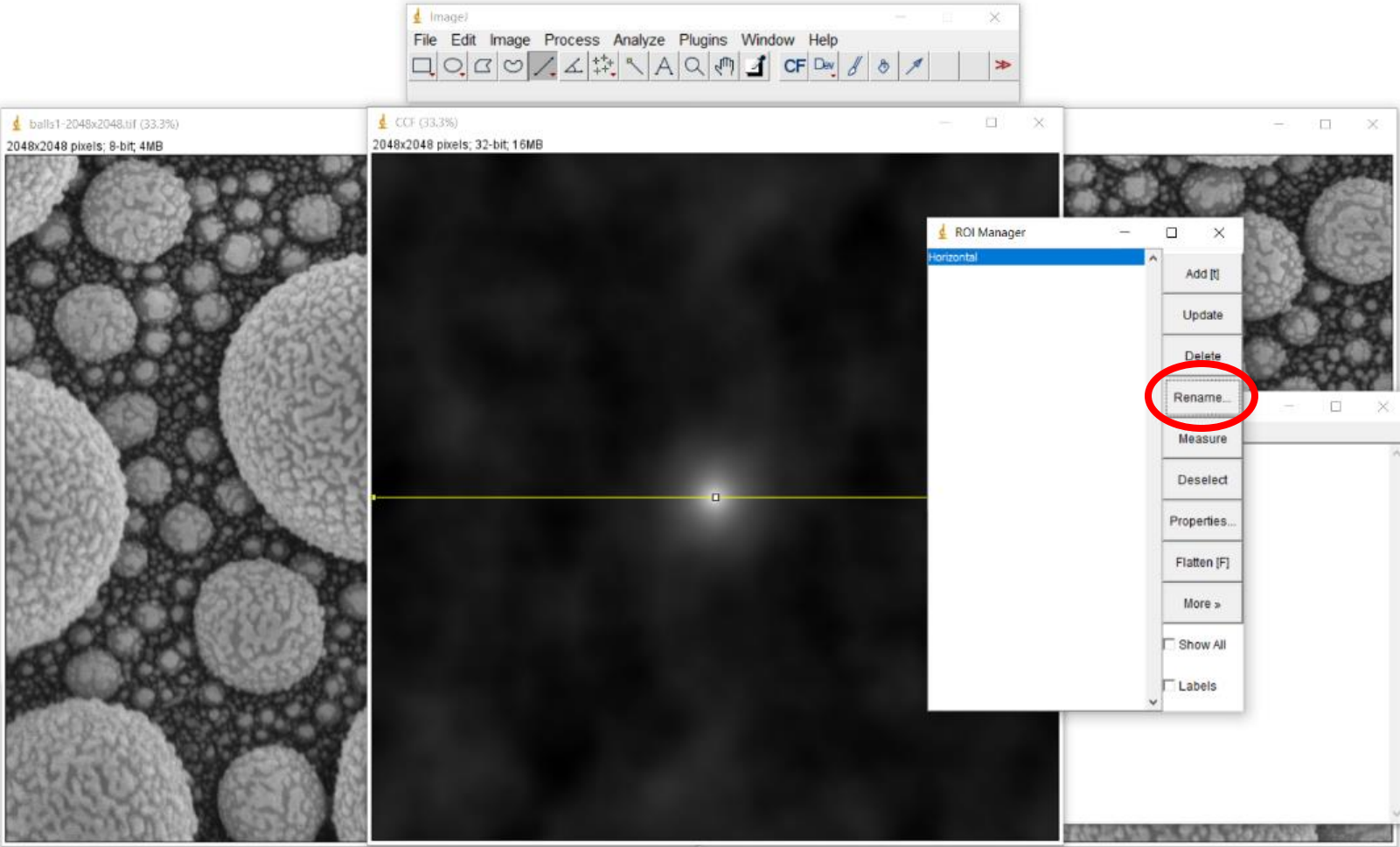
Go to Macros > Run Macro



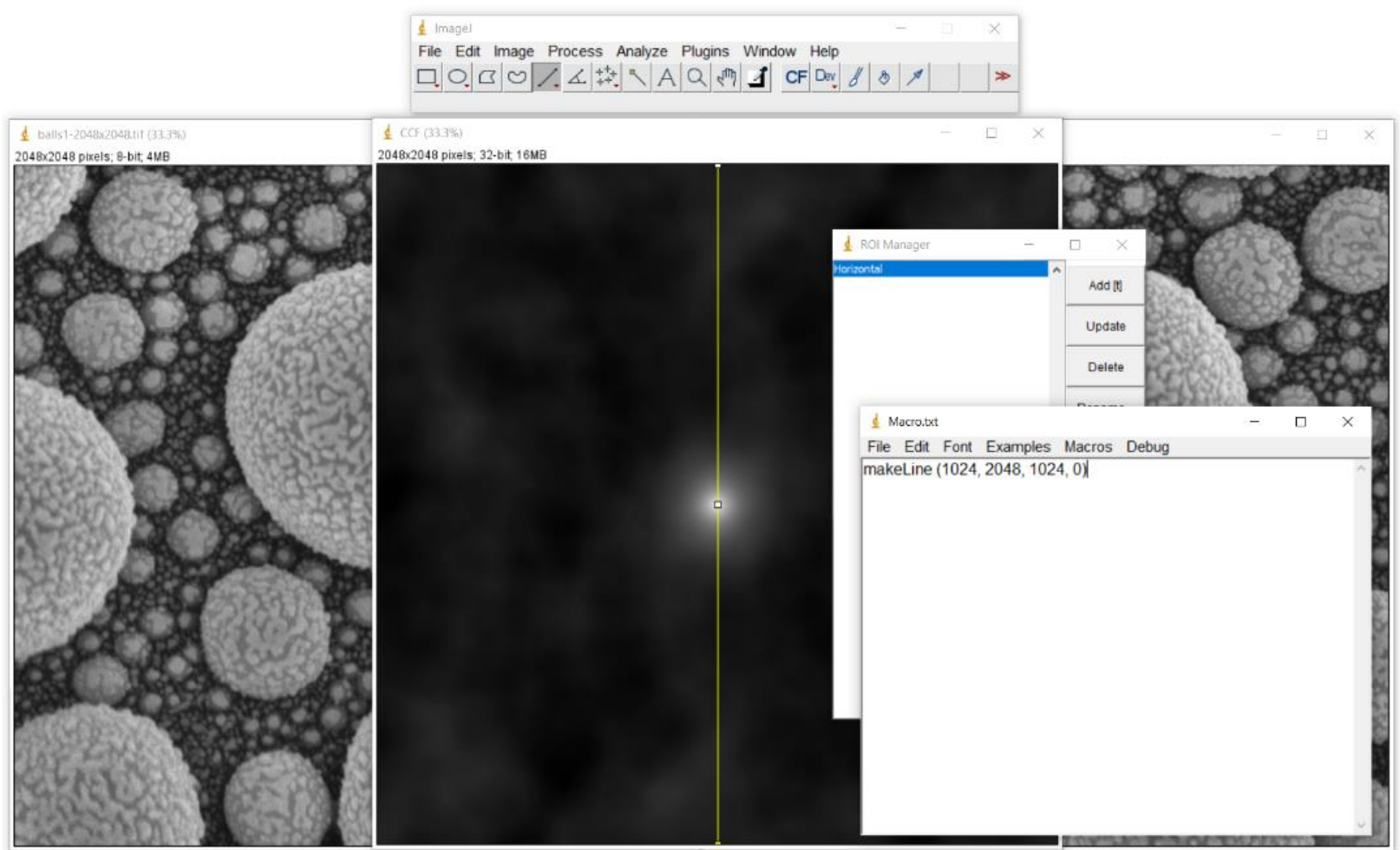
The result is a line with the perfect coordinates



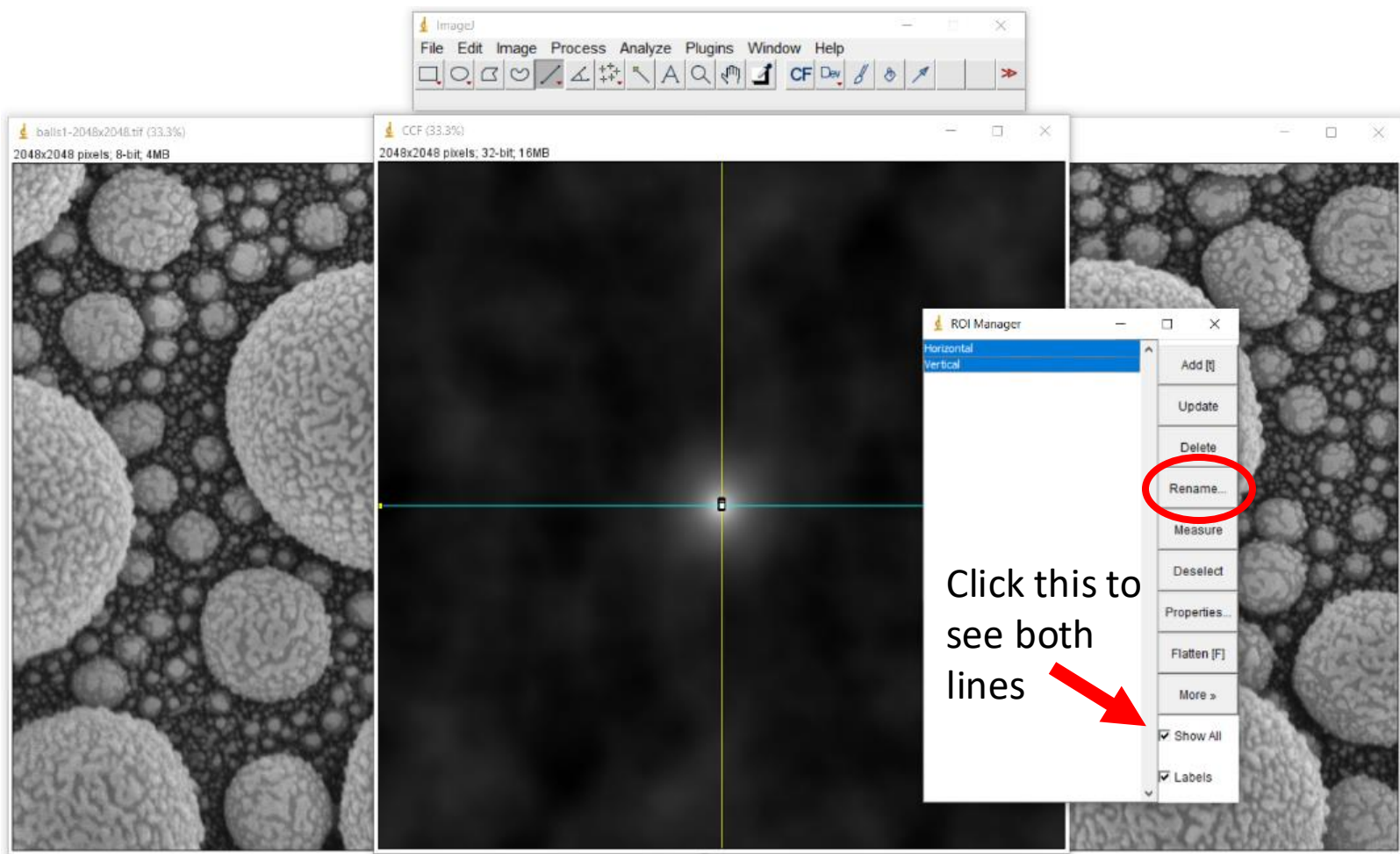
Go to your CCF and press Ctrl+t to add this line to the ROI manager, rename this to horizontal



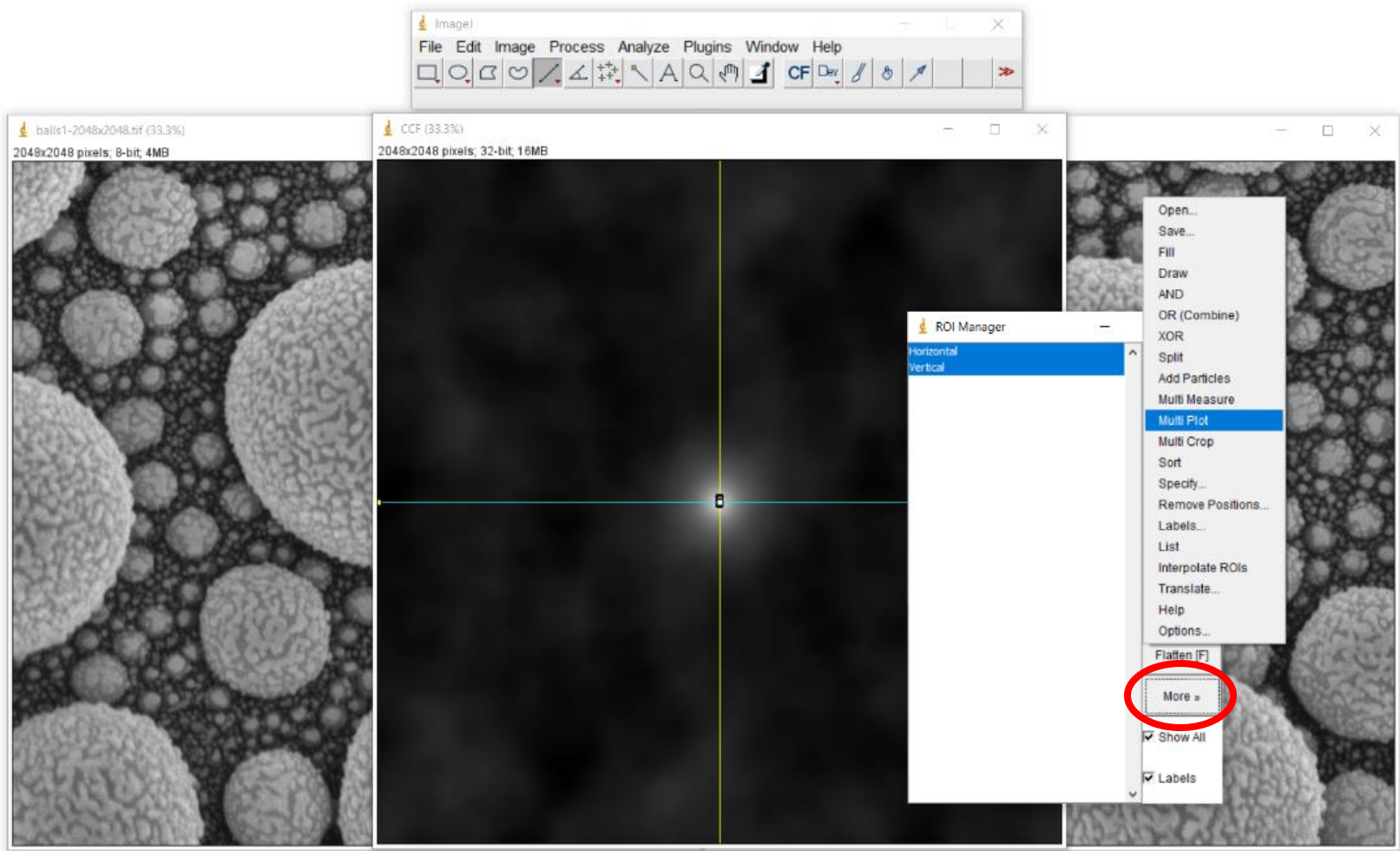
Go back to your macro and change the coordinates to (1024, 2048, 1024, 0), run it again



Go back to your CCF and press Ctrl+t to add the vertical line to the ROI manager, rename it



To plot both lines at the same time, select both by holding shift, click on More > Multi Plot



This will give you a plot with both lines. Export this data or capture an image

